

Asymptotic enumeration and inversion for A122399

A proof and reproducible research report

1 October 2026

Abstract

For $a_n = \sum_k k^n k! S(n, k)$, we prove the leading equivalent already recorded in OEIS and obtain a complete fixed-order expansion with explicit coefficients. An exact vertical contour gives a branch-safe representation and a simple, nonasymptotic exponential truncation bound. The same method yields compact-direction two-size asymptotics, Gaussian block-count fluctuations, and a controlled inverse with integer-threshold envelopes. An elementary appendix proves the posted mod-prime periodicity conjecture, in the sense that $p - 1$ is a period, and extends it to prime powers. The standard saddle methodology and existing leading result are explicitly credited.

[write] **Added 5 October 2026, batch 98: two Parts.** Since batch 98 this report has two Parts. Part I, from *Results and attribution* to *Sources and reproducibility*, is the batch-77 manuscript described in its provenance note; its text, labels and numbers are unchanged apart from dated notes. Part II (Part II, Sections 9–21 and Appendices A–B) adds manuscript 11 of batch 98, *From Gaussian Blocks to Poisson Defects*: an exact independent-Bernoulli representation of the number of missing blocks, finite Poisson and count bounds for all m, n , a degenerate, Poisson and Gaussian split at $n = m(\log m + c)$, uniform local Edgeworth expansions, a complete expansion in that window, an inverse for the singleton probability, and a local limit theorem on the diagonal. It answers Part I's Questions 2 and 4 in part. Its labels carry the prefix **a122:pd:**, and it reuses several letters of Part I with other meanings (its reading table): in particular its λ is $(m/2)e^{-n/m}$, not m/n , and its D is the number of missing blocks, not the growth constant.

Part I

Asymptotic enumeration and inversion for A122399

Results and attribution

The leading equivalent for A122399 was posted by Vaclav Kotesovec on OEIS on August 9, 2018; its growth constant was already identified in 2013. **Neither that equivalent nor its numerical constant is claimed as a new conjecture.** This manuscript proves it and develops explicit, reproducible higher coefficients, a globally valid contour with an explicit exponentially small truncation bound, a compact-direction extension, a block-count central limit theorem, and controlled inversion. These are applications and refinements of standard analytic saddle methods, not a new general method. In particular Khera–Lundberg–Melczer (2021, preprint 2019), Section 3 and the paragraph following its equation (15), give directional asymptotics and explain higher-order expansion for the nearby but different lonesum-matrix denominator. No global novelty claim is made.

[write] **Added 7 October 2026, batch 108: a neighbouring diagonal.** The collection’s report `a299907-lonesum-decomposable-matrices` (bundle Report 132) treats the diagonal of Kamano’s bivariate exponential generating function $\exp(x + y + 1/(1 - (e^x - 1)(e^y - 1)) - 1)$, which counts lonesum decomposable matrices (OEIS A299907). It works at the same $(n!)^2$ scale, with Stirling-number sums and a Lambert- W threshold inverse, but its singularity is the exponential of a pole on $(e^x - 1)(e^y - 1) = 1$, not a pole: the growth is $(n!)^2(\log 2)^{-2n}n^{-5/4}e^{\sqrt{2n/\log 2}}$ with an expansion in powers of $n^{-1/2}$, from a two-scale saddle (widths $n^{-3/4}$ and $n^{-1/2}$) instead of the single-scale saddle of (3.5). It also credits Khera–Lundberg–Melczer, for ordinary lonesum matrices. The two reports share no theorem, and nothing here is changed by it.

Provenance, status and reading conventions

[write] **Added 2 October 2026, batch 77. Provenance.** This report was built from one manuscript: batch 77, manuscript 35 of the repository’s incoming-reports channel, archive `oeis-a122399-research.zip` (main file `a122399-report.tex`, 389 lines, with an eight-page PDF that is not shipped), arrival commit `096ee7b87`, placed in `f76fcb566`. The text from the abstract to the end of the source list is the delivered manuscript, unchanged except for the label prefix `a122:` and the notes marked [write]. The delivered pre-layout source `proof.md` is shipped beside this article; it carries no mathematics absent here, while Section 8 exists only in this text. *Pin.* The delivery names no ProveIt commit that it continues. Its `proof.md` (line 9) records a scoped search of the public repository index on 1 October 2026, which found no A122399 or A317855 match and returned other files at commit `76c0887a6`; the delivered layout script removed that paragraph from the manuscript. The search result was true then; the only match now is this report. The search is scoped nonduplication evidence, as its author says, not a proof of absence. *Status.* An unrefereed research manuscript; the delivery names no author or tool. Nothing in it is formalized in Lean or Rocq, and its placement in the collection confers no formal status (Section 7 names the one formalized identity its proof uses). The delivered independent audit (shipped as `audit-independent_audit.md`) found no blocking error; its script was not replayed at placement (it exceeded a 170-second cap), while the four producer scripts were replayed with identical outputs (up to line endings).

[write] **Reading conventions.** The manuscript reuses several letters. No symbol of the delivered text was renamed; the table fixes the readings.

symbol	meanings, by section
p	$p(z) = 1/(1 + e^z)$ in Sections 2–4; $p = x/y$ in the formula for μ_0 and $p(s)$ in Section 6; a prime in Section 7.
r	the Taylor index of f_r in (3.2), (3.4); the Newton step count in (5.3); the prime-power exponent in (7.1); the OEIS constant $r = 0.873702\dots$, which is $\mu = 1/y$ of (6.2), not any of the others.
x, y	formal variables of $F(x, y; u)$ in Section 1; saddle values $x = X(y)$, $y = y(\lambda)$ from (3.1) on, and on the diagonal the numbers $x(1)$, $y(1)$ of Section 4.
a	a_n is the sequence; a is a summation index in (3.4) and $a = y - x$ in (4.2).
m	the red set size (with $\lambda = m/n$); m_r in (3.4) are multiplicities.
C, C_J, E_J	C is the diagonal constant of (4.1); C_J and E_J in the proof of (5.4) are unrelated existence constants; $C(s)$ in Section 6 has $C(0) = C$.
μ, μ_0	mean of the block count (6.2), not a growth rate.
D	the growth constant of (4.1): $D = 1/(xy) = (1 + e^y)/y^2 = 3.16108865\dots$, the constant of OEIS A317855.

Section and equation numbers are the manuscript’s own and are typed by hand; equation tags such as (2.1) are fixed tags, not counters.

1. Exact combinatorics

Put

$$A_{m,n} = \sum_{k=0}^m k^n k! S(m, k), \quad a_n = A_{n,n}, \quad b_{m,n} = \frac{A_{m,n}}{m!n!}.$$

The convention at the origin is $0^0 = 1$. For positive m, n , a structure consists of an ordered partition of an m -element red set into k nonempty blocks, followed by a map of an n -element blue set to these blocks. Its bivariate exponential generating function, with u marking blocks, is

$$F(x, y; u) = \frac{1}{1 - ue^y(e^x - 1)}.$$

Thus the diagonal is exactly A122399, not an approximation to it. When a structure is chosen uniformly, its block count satisfies $\Pr(K_n = k) = k^n k! S(n, k) / a_n$.

2. Exact contour and a nonasymptotic tail bound

For $\Re z > 0$, use the single analytic branch

$$X(z) = \text{Log}(1 + e^{-z}), \quad p(z) = \frac{1}{1 + e^z} = -X'(z).$$

Here $1 + e^{-z}$ stays in the open disk of radius one about 1, and $X(z) \neq 0$. The following identities hold for every pair of integers $m, n \geq 1$ and every $c > 0$:

$$\begin{aligned} b_{m,n} &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{p(z)}{X(z)^{m+1} z^{n+1}} dz \\ &= \frac{n+1}{m} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} X(z)^{-m} z^{-n-2} dz. \end{aligned} \tag{2.1}$$

Both integrals are absolutely convergent. Integer powers are used, so there is no unresolved power branch.

Proof

For $\Re h > 0$ and $n \geq 1$, the absolutely and locally uniformly convergent Mittag-Leffler identity is

$$\frac{1}{n!} \sum_{k \geq 1} k^n e^{-kh} = \sum_{j \in \mathbb{Z}} (h + 2\pi i j)^{-n-1}. \tag{2.2}$$

It follows, for example, by differentiating $1/(e^h - 1)$'s symmetric partial-fraction expansion; the constant term disappears after one derivative. It also follows by Poisson summation for $t^n e^{-ht} \mathbf{1}_{t \geq 0} / n!$, whose Fourier coefficients are the terms on the right.

Change the x -coefficient contour to $q = e^x - 1$, with $|q| = e^{-c} < 1$; the inverse is $x = \text{Log}(1 + q)$, analytic and one-to-one near the closed contour, and it winds once around zero. On this circle choose the continuous parametrization $q = e^{-c+it}$, $-\pi \leq t \leq \pi$, so $h = c - it$. Insert (2.2) in the Cauchy coefficient integral. Uniform absolute convergence permits the sum-integral interchange. The shifted intervals in $h + 2\pi i j$ join to the whole vertical line; X and p are $2\pi i$ -periodic. This gives the first identity in (2.1). Since $(X^{-m})' = mpX^{-m-1}$, integration by parts gives the second. Its boundary term tends to zero: on a fixed vertical line X is bounded away from zero, and $z^{-n-1} \rightarrow 0$. This also proves absolute convergence.

This argument does **not** treat an individual $(h(x) + 2\pi i j)^{-n-1}$ as a globally analytic power series at $x = 0$. Doing so would ignore its logarithmic monodromy.

Explicit truncation theorem

Let $x_c = X(c) > 0$, and let R_T denote the sum of the parts of the **second** integral in (2.1) with $|\Im z| \geq T > 0$, including its prefactor. Then

$$|R_T| \leq \frac{x_c^{-m} c^{-n}}{\pi m T} \left(\frac{c}{\sqrt{c^2 + T^2}} \right)^n. \quad (2.3)$$

Indeed, with $w = X(c + it)$,

$$e^{-c} = |e^w - 1| \leq e^{|w|} - 1, \quad \text{so} \quad |X(c + it)| \geq X(c).$$

Consequently the modulus of the tail is bounded by $(n+1)x_c^{-m}/(\pi m) \int_T^\infty (c^2 + t^2)^{-(n+2)/2} dt$. Set $v = \sqrt{c^2 + t^2}$; since $v/\sqrt{v^2 - c^2}$ decreases, that integral is at most $(c^2 + T^2)^{-n/2}/((n+1)T)$, proving (2.3).

There is therefore an exact, absolutely convergent strip decomposition obtained by splitting the second contour into $(2j-1)\pi \leq \Im z \leq (2j+1)\pi$, $j \in \mathbb{Z}$. The noncentral strips together have bound (2.3) with $T = \pi$. They are not asserted to be canonical saddle/transseries sectors: their artificial endpoint contributions can cancel, and no individual secondary-saddle expansion is proved here. A fixed-order Poincaré expansion for the central strip has algebraic error, despite the exponentially accurate contour truncation.

3. The unique saddle and all orders

Fix $\lambda = m/n > 0$. There is a unique $y = y(\lambda) > 0$ satisfying

$$x = X(y) = \lambda y p(y), \quad B = B(\lambda) = 1 + \lambda^{-1} - y(1 - p(y)). \quad (3.1)$$

Moreover $B > 1$, and x, y, B depend analytically on λ . To prove uniqueness, $R(y) = yp(y)/X(y)$ increases strictly from 0 to infinity. In fact $p/X > 1 - p$, since $p + (1-p)\log(1-p) > 0$ for $0 < p < 1$; differentiating $\log(p/X)$ shows its derivative is positive. Thus $R' > 0$. At the root $y(1-p) < 1/\lambda$, proving $B > 1$. The implicit-function theorem supplies analyticity.

Define the analytic phase germ at $s = 0$

$$f_\lambda(s) = -\log(1+s) - \lambda \log \frac{X(y(1+s))}{x} = \frac{B}{2}s^2 + \sum_{r \geq 3} f_r s^r. \quad (3.2)$$

Here logarithms are local germs, and $f_\lambda(0) = f'_\lambda(0) = 0$. Globally along $s = i\theta$, the exponential in the following formula means its integer-power expression, so no global choice of f is needed:

$$b_{m,n} = \frac{x^{-m} y^{-n}}{\lambda y} \left(1 + \frac{1}{n} \right) \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{nf_\lambda(i\theta)}}{(1+i\theta)^2} d\theta. \quad (3.3)$$

The bound from Section 2 gives $|e^{nf_\lambda(i\theta)}| \leq (1 + \theta^2)^{-n/2}$.

Explicit coefficient formula

Set $(-1)!! = 1$. For $j \geq 0$, sum over nonnegative integers $a, m_3, \dots, m_{2j+2}$ subject to

$$a + \sum_{r=3}^{2j+2} (r-2)m_r = 2j, \quad M = a + \sum_{r=3}^{2j+2} r m_r.$$

The integer M is necessarily even. Define

$$d_j = \sum (-1)^a (a+1) \prod_{r=3}^{2j+2} \frac{f_r^{m_r}}{m_r!} \frac{(-1)^{M/2} (M-1)!!}{B^{M/2}}, \quad c_0 = 1, \quad c_j = d_j + d_{j-1} \quad (j \geq 1). \quad (3.4)$$

The empty product gives $d_0 = 1$. For every fixed integer $J \geq 0$,

$$A_{m,n} = \frac{m!n! x^{-m} y^{-n}}{\lambda y \sqrt{2\pi n B}} \left(\sum_{j=0}^J \frac{c_j(\lambda)}{n^j} + O(n^{-J-1}) \right). \quad (3.5)$$

The remainder is uniform when m/n ranges over any fixed compact subset of $(0, \infty)$. No growing-direction or boundary-uniform assertion is made.

[write] Added 5 October 2026, batch 98. This remains true of (3.5). Part II treats the boundary $m/n \rightarrow 0$ by a different route: finite bounds for every m, n (Theorem 12.1), the degenerate, Poisson and Gaussian regimes for every $n(m)$ (Theorem 13.1), and a complete expansion of $A_{m,n}$ in the window $n = m(\log m + O(1))$ (Theorem 15.1). On the side $m/n \rightarrow \infty$ it proves only distributional statements. No approximation joining that window to (3.5) is known (Question 20.3). Note that Part II's λ is $(m/2)e^{-n/m}$, not the $\lambda = m/n$ used here.

All-orders proof and remainder justification

Choose a small fixed $\delta > 0$ uniformly over the compact direction set. Outside $|\theta| < \delta$, the modulus bound above makes the integral exponentially small; integrate the power tail exactly or use (2.3). Inside, analyticity, $B > 1$, and compactness give uniform Taylor remainder bounds and a uniform quadratic bound $\Re f(i\theta) \leq -b\theta^2$, for some $b > 0$. Set $\theta = t/\sqrt{n}$. Expand the analytic amplitude $(1+s)^{-2}$ and $\exp\{n(f(s) - Bs^2/2)\}$ through order $n^{-J-1/2}$, keeping enough further Taylor terms to bound the remainder. On $|t| \leq n^\epsilon$, with $\epsilon > 0$ chosen sufficiently small for this fixed J , Taylor's theorem bounds the remainder by n^{-J-1} times a fixed polynomial in $|t|$ times a Gaussian with uniformly negative exponent. On $n^\epsilon < |t| < \delta\sqrt{n}$, the uniform quadratic bound is superpolynomially small. Extend the polynomial–Gaussian integrals to the real line with the same accuracy. Odd terms integrate to zero. The even Gaussian moments are $(M-1)!!B^{-M/2}$, with $i^M = (-1)^{M/2}$, giving (3.4). The prefactor $1 + 1/n$ gives $c_j = d_j + d_{j-1}$. This proves (3.5) with its uniform remainder.

This is the ordinary analytic one-dimensional saddle expansion; the exact contour is what supplies the particularly simple global tail control here.

4. Diagonal constants and reproducible coefficients

On the diagonal write $x = x(1)$, $y = y(1)$, $B = 2 - y + x$, $D = (xy)^{-1}$, and

$$C = \frac{1}{y\sqrt{2\pi B}}.$$

Numerically,

- $x = 0.2763928914715160816083833582415360977661$
- $y = 1.144554440911905470205106411159417937429$
- $B = 1.131838450559610611403276947082118160337$
- $D = 3.161088653865428813830172202588132491726$
- $C = 0.3276282855698694814422864924105070307103$

Thus

$$a_n = \frac{CD^n(n!)^2}{\sqrt{n}} \left[1 - \frac{0.0600744155748035800316371200380}{n} + \frac{0.00759452281787645507680871266458}{n^2} + \frac{0.00371322633571244961947936026057}{n^3} - \frac{0.000362980958541489832391365216986}{n^4} + O(n^{-5}) \right]. \quad (4.1)$$

An exact expression for the first coefficient is

$$c_1 = 1 - \frac{3}{B} + \frac{3f_4 - 6f_3}{B^2} - \frac{15f_3^2}{2B^3}, \quad (4.2)$$

where, setting $a = y - x$, $b = y - 2x$,

$$f_3 = \frac{a(b-3)}{6}, \quad f_4 = \frac{1}{2} - \frac{a}{2} + \frac{a^2}{8} + \frac{ab}{6} - \frac{a(y^2 - 6xy + 6x^2)}{24}.$$

The numerical generator uses exact rational symbolic algebra for the phase, then 90-digit arithmetic; it does not fit these coefficients to sequence values. It prints coefficients through n^{-7} , exact checks through $n = 800$, and scaled remainders. The recurrence $T_{m,k} = k(T_{m-1,k} + T_{m-1,k-1})$, $T_{0,0} = 1$, generates $T_{m,k} = k!S(m,k)$, so the numerical checks use independent exact integer values.

The strip-tail factor for $T = \pi$ is $\rho_\pi = y/\sqrt{y^2 + \pi^2} < 0.343$. Accordingly the exact central-contour integral differs from $a_n/(n!)^2$ by at most $D^n \rho_\pi^n / (\pi^2 n)$. This is a contour-evaluation statement, not an exponentially accurate bound for a fixed truncation of (4.1).

5. Controlled inverse

Define $P_J(t) = \sum_{j=0}^J c_j t^j$. For every fixed J , it is positive for all sufficiently small nonnegative t . For sufficiently large real v , put

$$H_J(v) = 2 \log \Gamma(v+1) + v \log D - \frac{1}{2} \log v + \log C + \log P_J(1/v).$$

Then

$$\log a_n = H_J(n) + O(n^{-J-1}), \quad H'_J(v) = 2 \log v + \log D + O(v^{-1}) > 0.$$

Consequently the large solution ν_J of $H_J(\nu_J) = \log a_n$ is unique and satisfies

$$\boxed{\nu_J = n + O(n^{-J-1}/\log n)}. \quad (5.1)$$

This follows from the mean-value theorem, using (3.5) and the derivative estimate. It is a controlled inverse along the sequence, without choosing an arbitrary exact interpolation of the integers.

For a practical Lambert-W initialization, put $L = \log A$ and

$$w = W_0 \left(\frac{\sqrt{D}}{2e} L \right), \quad N_0 = \frac{L}{2w}.$$

If $A = a_n$, then $N_0 = n + O(1)$, and

$$n = N_0 - \frac{\frac{1}{2} \log N_0 + \log(2\pi C)}{2(w+1)} + O\left(\frac{1}{N_0 \log N_0}\right). \quad (5.2)$$

Indeed Stirling's expansion gives $\log a_n = 2n \log n + (\log D - 2)n + \frac{1}{2} \log n + \log(2\pi C) + O(1/n)$, while N_0 exactly solves its first two terms. Taylor expansion around N_0 gives (5.2).

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The initializer of (5.2) is an instance of the inverse-Gamma core of the transseries-and-inversion volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, section `p6:sec:gamma`), and no novelty is claimed for the method. The two leading Stirling terms read $2N \log N + (\log D - 2)N = L$; with $T = \sqrt{D} N$ and $R = \sqrt{D} L/2$ this is the core equation $T(\log T - 1) = R$ of `p6:eq:gamma-core-eq`, whose solution $w = W_0(R/e)$, $T = R/w$ is $w = W_0(\sqrt{DL}/(2e))$, $N_0 = L/(2w)$ as above. The general power-logarithmic block is `p6:lem:core`, and the scaled family is `p2:thm:multi-scaling`. Only the core is shared: the lower-order terms here, $\frac{1}{2} \log N + \log(2\pi C)$, are not those of $\log \Gamma$, so the volume's all-orders recurrence for Γ_+^{-1} does not apply as it stands; this report's own all-orders inverse is ν_J of (5.1).

For arbitrary algebraic inverse precision use Newton iteration

$$N_{r+1} = N_r - \frac{H_J(N_r) - L}{H_J'(N_r)}, \quad (5.3)$$

starting at N_0 . Uniformly in its $O(1)$ starting neighborhood, $H_J''/H_J' = O((n \log n)^{-1})$. Therefore its error relative to ν_J after fixed r iterations is $O((n \log n)^{-(2^r-1)})$. Taking $2^r - 1 \geq J + 1$ and combining with (5.1) gives the inverse error in (5.1).

For an arbitrary large target A , let $\nu = H_J^{-1}(\log A)$ and $M(A) = \min\{n : a_n \geq A\}$. There is a constant $E_J > 0$ such that

$$\left| \nu - E_J \frac{\nu^{-J-1}}{\log \nu} \right| \leq M(A) \leq \left| \nu + E_J \frac{\nu^{-J-1}}{\log \nu} \right| \quad (5.4)$$

for all sufficiently large A . Indeed, the uniform sequence-value error $|\log a_k - H_J(k)| \leq C_J k^{-J-1}$ and the lower bound on H_J' show that integers below the lower real endpoint have $a_k < A$, and an integer at or above the upper endpoint has $a_k \geq A$. Enlarge E_J to cover both comparisons. The sequence is strictly increasing from $n = 1$ onward: its old positive block-count summands weakly increase and additional positive structures occur.

The envelope (5.4) allows two neighboring candidates near an integer boundary, so rounding a real inverse alone is not guaranteed. Exact integer comparison settles that boundary. The constants and starting threshold in (5.4) are asymptotic existence constants, not a certified finite-input error bar unless separately bounded.

[write] **Added 2 October 2026, batch 77.** The rounding remarks above are the general staircase facts of the same volume, `p0:thm:staircase`: part (1), $M(A) = \lceil \nu \rceil$ for the exact monotone interpolation, and part (2), the separation condition that fails near an integer boundary. Their generic arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`); those statements concern an arbitrary strictly monotone function and give no formal status to (5.1)–(5.4), whose interpolation H_J is not exact.

6. Block-count fluctuations

For $u = e^s$ near 1, replace X by $X_s(z) = \text{Log}(1 + e^{-s-z})$, and let $y(s)$, $x(s) = X_s(y(s))$ solve $x(s) = y(s)p(s)$, where $p(s) = 1/(1 + e^{s+y(s)})$. Set

$$B(s) = 2 - y(s)(1 - p(s)), \quad D(s) = [x(s)y(s)]^{-1}, \quad C(s) = \frac{1}{y(s)\sqrt{2\pi B(s)}}.$$

The all-orders theorem extends analytically and uniformly to a sufficiently small complex disk in s . To justify this rather than assuming it, use the analytic implicit-function theorem at the nondegenerate saddle. The contour can be locally deformed through $y(s)$ inside $\Re(s + z) > 0$. The fixed outer contour portions retain a strict exponential gap by continuity from $s = 0$, and the far tails retain uniform power bounds. The local analytic saddle expansion is therefore uniform in complex s . Cauchy estimates allow differentiation of its remainder.

It follows that

$$\mathbb{E}e^{sK_n} = \frac{C(s)}{C(0)} \left(\frac{D(s)}{D(0)} \right)^n [1 + O(n^{-1})], \quad (6.1)$$

uniformly near zero. This is the standard analytic quasi-powers situation. A direct expansion of the logarithm of (6.1) at $s = it/\sqrt{n}$, followed by the continuity theorem, gives the Gaussian limit stated below.

Let $\mu = 1/y$ and $\sigma^2 = (B - 1)/(By^2) > 0$. Then

$$\begin{aligned} \mathbb{E}K_n &= \mu n + \mu_0 + O(n^{-1}), & \frac{K_n - \mu n}{\sigma\sqrt{n}} &\Longrightarrow N(0, 1), \\ \text{Var}(K_n) &= \sigma^2 n + O(1), \end{aligned} \quad (6.2)$$

where

$$\mu_0 = \frac{B - 1}{By} - \frac{B - 1 - p}{2B^2}, \quad p = x/y.$$

For verification, envelope differentiation gives $(\log D)' = 1/y$. Implicit differentiation of the saddle equation gives $y' = 1/B - 1$, hence $(\log D)'' = (B - 1)/(By^2)$. Further, $B' = (B - 1 - p)/B$ and $(\log C)' = -y'/y - B'/(2B)$, yielding μ_0 .

Numerically,

$$\begin{aligned} \mu &= 0.873702433239668330496568304721, \\ \sigma^2 &= 0.0889169868271170325484054553784, \\ \mu_0 &= 0.144565688577761026837729354227. \end{aligned}$$

This gives a probabilistic interpretation of the OEIS growth parametrization's constant $r = 0.873702\dots$: it is the asymptotic fraction of occupied ordered blocks. No local limit theorem or growing-marking-range result is claimed.

[write] Added 5 October 2026, batch 98. Part II proves the local limit theorem on the diagonal, $\Pr(K_n = k) = (\sigma\sqrt{n})^{-1}\varphi((k - \mu n - \mu_0)/(\sigma\sqrt{n})) + O(n^{-1})$ uniformly in k , with φ the standard normal density (Corollary 18.1), from an exact independent-Bernoulli representation of $n - K_n$ and a uniform lattice Edgeworth expansion at exact cumulants (Theorem 14.1); the constants μ, σ^2, μ_0 are those of (6.2). No growing-marking-range result and no large-deviation estimate is proved there either.

7. Elementary congruence appendix

The current OEIS entry conjectures that, for a prime p , the reduction of a_n for $n \geq 1$ is purely periodic with period $p - 1$. The following proves that $p - 1$ is a period; it does not assert that it is always the least period.

Proposition. For every prime p and integer $r \geq 1$,

$$a_{n+\varphi(p^r)} \equiv a_n \pmod{p^r} \quad (n \geq r). \quad (7.1)$$

In particular $a_{n+p-1} \equiv a_n \pmod{p}$ for every $n \geq 1$.

Proof. Terms with $k \geq pr$ vanish modulo p^r , since $v_p(k!) \geq \lfloor k/p \rfloor \geq r$. We may extend the sum up to $pr - 1$ even when this exceeds n , since $S(n, k) = 0$ for $k > n$. The exact inclusion-exclusion formula gives

$$a_n \equiv \sum_{k=0}^{pr-1} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (kj)^n \pmod{p^r}.$$

For $n \geq r$, a term with $p \mid kj$ is zero modulo p^r . Each other base kj is a unit, so Euler's theorem leaves its power unchanged on replacing n by $n + \varphi(p^r)$. This proves (7.1). The argument is elementary and independent of the asymptotic analysis.

[write] **Added 2 October 2026, batch 77.** The OEIS conjecture is credited there to Peter Bala (31 May 2022), as the delivered `source_review.md` records. Neither (7.1) nor any other statement of this report is formalized. The one nontrivial identity in its proof, the surjection formula $k!S(n, k) = \sum_{j \leq k} (-1)^{k-j} \binom{k}{j} j^n$, is formalized in Lean as `Fabius.factorial_mul_stirlingSecond_eq_sum` (`Analysis/FabiusFunction/Lean/FabiusFunction/StirlingBasisChange.lean`), so (7.1) is a small formalization target on top of it and Euler's theorem; no such proof exists in the repository. The collection's report `congruences-and-valuations/secant-number-periodicity` studies the same kind of question for the secant numbers A000364, where the analogous OEIS conjecture (period dividing $\varphi(m)$, pure from index one) is false for some moduli; no result is shared, and no least period or onset is claimed here.

8. Questions left by this analysis

The results separate three different levels of control: an exact contour with an explicit exponential tail, a fixed-order algebraic saddle expansion, and an asymptotic inverse. Several natural extensions require additional arguments.

1. **Exponentially improved expansions.** Locate and classify the complex saddles of $z \operatorname{Log}(1 + e^{-z})$, determine which are reached by valid contour deformations, and obtain an optimally truncated expansion with a controlled secondary-saddle remainder. The exact strip decomposition alone does not answer this question.
2. **Directions approaching the boundary.** Determine uniform laws when $m/n \rightarrow 0$ or $m/n \rightarrow \infty$, including the transition scales where the compact-direction expansion ceases to be uniform. These regimes may require different saddle coordinates and block-count normalizations.

[write] **Added 5 October 2026, batch 98: answered in part.** For $m/n \rightarrow 0$, Part II locates the transition at $n = m(\log m + c)$, where the number of missing blocks tends to a Poisson law of parameter $e^{-c}/2$ and $A_{m,n}/(m!m^n) \rightarrow \exp(e^{-c}/2)$; it proves a complete expansion there (Theorem 15.1), explicit bounds for all m, n (Theorem 12.1), and the regime of every path $n(m)$ (Theorem 13.1). For $m/n \rightarrow \infty$ it gives Gaussian local laws at exact moments (Theorem 14.1) but no count asymptotics. Open: count asymptotics for $m/n \rightarrow \infty$, and a uniform count approximation joining the window, the growing Poisson region and (3.5) (Question 20.3).

3. **Effective finite-input inverse bounds.** Turn the implicit all-orders constants into computable numerical bounds and starting thresholds, combining analytic Taylor estimates with the explicit contour tail. This would make the ceiling envelope a certified finite-input bracket and could reduce the need for exact comparisons away from sequence-value boundaries.

4. **Sharper block-count laws.** Prove a local limit theorem, explicit Edgeworth terms, or a uniform large-deviation estimate for the block count. The analytic marking germ provides the central cumulants, but the required global Fourier or tilted-contour estimates are not supplied here.

[write] **Added 5 October 2026, batch 98: answered in part.** The local limit theorem is proved in Part II (Corollary 18.1), with error $O(n^{-1})$, and Edgeworth terms to every fixed order are proved at the exact cumulants for every path with diverging variance (Theorem 14.1); the global Fourier estimate comes from the real-rootedness of the block polynomial, not from a contour. Open: the Edgeworth terms in fixed diagonal constants (Question 20.8) and any large-deviation estimate.

These are limitations and possible continuations of this analysis, not claims that the corresponding general techniques or all related results are absent from the literature.

Sources and reproducibility

1. OEIS A122399, current entry inspected 2026-10-01: <https://oeis.org/A122399>
2. OEIS A317855: <https://oeis.org/A317855>
3. Jessica Khera, Erik Lundberg, Stephen Melczer, *Asymptotic Enumeration of Lonesum Matrices*, Advances in Applied Mathematics 123 (2021), article 102118, doi:10.1016/j.aam.2020.102118; arXiv:1912.08850: <https://arxiv.org/html/1912.08850> (especially Section 3, Theorem 3.1, and higher-order comment after equation (15)). Its denominator is $e^{-x} + e^{-y} - 1$, different from the one here.
4. `check_a122399.py`: exact rational phase generation, finite-sum saddle coefficients, exact-integer comparisons, and moment checks; outputs `numerical_results.json`, `phase_coefficients.txt`, and `test_output.txt`.
5. `export_exact_coefficients.py`: exact rational expressions through the fourth correction, including fully combined rational functions of x, y for the first two corrections; the exporter asserts that no floating-point atoms occur.
6. `verify_contour_inverse.py`: independent direct quadrature of the finite vertical contour with the explicit tail certificate, and inverse-iteration checks.
7. `verify_congruences.py`: finite modular checks of (7.1), including all primes below 50 and several prime powers.

[write] **Added 2 October 2026, batch 77.** *Item 2.* The manuscript lists OEIS A317855 without saying what it is. Checked on 2 October 2026: A317855 is the decimal expansion $3.161088653\dots$ of the growth constant D of (4.1) (OEIS titles it as a constant of the asymptotics of A122400), and the A122399 entry writes the Kotěšovec equivalent as $c \cdot A317855^n (n!)^2 / \sqrt{n}$ with $c = 0.32762828\dots = C$, its constant r being $1/y$. The manuscript states no other relation between A317855 and A122399. *Items 4–7 and the audit.* The scripts are shipped as `code/check_a122399.py`, `code/export_exact_coefficients.py`, `code/verify_contour_inverse.py` and `code/verify_congruences.py`, their recorded outputs under `data/`, and the audit as `audit-independent_audit.md`, `code/audit-independent_audit.py` and `data/audit-independent_audit_*`. They read and write files next to themselves in the delivered flat layout; the report’s README gives rerun instructions that do not overwrite shipped data. *Neighbours.* Row $n = 0$ of $A_{m,n}$ is the Fubini sequence, $A_{m,0} = \sum_k k! S(m, k)$, defined in Lean as `Fabius.fubini` (`Analysis/FabiusFunction/Lean/FabiusFunction/OrderedBell.lean`); its exact pole-lattice formula is `q2:thm:fubini` of the transseries volume. That row lies outside the compact-direction theorem (3.5), on the side of question 2 of Section 8, and nothing is claimed about it here. The collection’s `a277364-bell-asymptotics` treats a different Stirling sum by classical saddle methods; no theorem is shared.

[write] **Added 5 October 2026, batch 98.** Part II covers the Fubini row $n = 0$ qualitatively: its block polynomial $\sum_k k! S(m, k) u^k$ has only real zeros, in $(-1, 0]$, the case $n = 0$ of Theorem 11.1, which is Frobenius’s theorem on the Eulerian polynomials after a change of variable (note in Section 9); and the number

of missing blocks is asymptotically Gaussian with the local laws of Theorem 14.1, since $\theta = (m - 1)/2 \rightarrow \infty$. No count asymptotics for that row are proved in either Part.

The computer checks are reproducible evidence; the all-orders and error statements rely on the analytic proofs above. An independent audit and clean replay accompany this report. Conventional peer review remains appropriate before publication.

Part II

From Gaussian blocks to Poisson defects

Part II: provenance, status and reading conventions

[write] Added 5 October 2026, batch 98. *Provenance.* Part II was added in the write of batch 98. It is built from one manuscript: batch 98, manuscript 11, the archive `A122399_Boundary_Transitions_and_Inversion.zip` (351,864 bytes), which arrived in commit `2172df76a` and was placed in this report by commit `9353a7171`. Its main file `article.tex` (1,480 lines) is titled *From Gaussian Blocks to Poisson Defects. Boundary transseries, local limit laws, and inverse thresholds for the rectangular family of OEIS A122399* and dated 4 October 2026; its author line reads “ChatGPT” and “Research continuation prepared for Vladimir Reshetnikov”. Its 22-page PDF, its delivery README and its checksum manifest are not shipped (the manifest was verified, 21 of 21 files, at placement). *Pin.* The manuscript records the tree `0f93381e8` (4 October 2026) and the blob `6f03a8be2` of this report’s `proof.md`, which is still the shipped blob; the report had no commit between the pin and the placement, so the manuscript read the current Part I. *What is printed.* Sections 9–21 and Appendices A–B below are the manuscript’s Sections 1–13 and Appendices A–B. Section k of the manuscript is Section $k + 8$ here, and every number inside it moves with it: the manuscript’s equation (4.10) is (12.10) here, its Theorem 9.1 is Theorem 17.1 and its Question 12.3 is Question 20.3; appendix numbers are unchanged. The text is the delivered text except for the label prefix `a122:pd:`, the symbol renames listed below, the shipped names of the two tables read by `\input`, the title of Section 20 (“Further research questions” in the manuscript), the corrected bibliography item [3], three bibliography items added by the write, and the notes marked [write]. The manuscript’s own abstract and status paragraph follow.

Abstract (manuscript 11). The diagonal sequence A122399 is $a_n = \sum_k k^n k! S(n, k)$. Its natural rectangular extension is $A_{m,n} = \sum_k k^n k! S(m, k)$, counting an ordered partition of an m -element set together with a map from an n -element set to its blocks. The existing ProveIt report develops asymptotics in fixed positive m/n directions and a diagonal central limit theorem, while explicitly excluding boundary-uniform asymptotics and local limit theorems. We address these exclusions by a different, finite-polynomial route. The block polynomial has a simple zero at zero and all its other zeros in $(-1, 0)$. Consequently the number of missing blocks is exactly a sum of independent Bernoulli variables. Two explicit coefficients then give nonasymptotic Poisson and enumeration bounds, and a Gaussian–Poisson–degenerate trichotomy with critical scale $n = m(\log m + c)$. In this critical window we prove a complete expansion in m^{-1} with polynomial logarithmic coefficients, uniformly for bounded complex marking variables, and give a finite rational algorithm for every coefficient. We also derive all-order local Edgeworth expansions whenever the variance diverges, and invert the critical expansion to locate a prescribed probability of the all-singleton partition. Exact symbolic checks and high-precision numerical diagnostics accompany the proofs. These are specific extensions of the repository’s report, not a claim of a new general asymptotic method or independently established worldwide priority.

Status (manuscript 11). Self-contained research manuscript, not independently refereed and not Lean- or Rocq-formalized. The main arguments are proofs; finite computer checks are supplementary evidence rather than substitutes for those proofs. No repository or OEIS entry has been modified.

[write] Added 5 October 2026, batch 98: **status, scope and relation to Part I.** *Status.* Unrefereed and not formalized; placement in the collection confers no formal status (the one formalized ingredient is named after Corollary 11.2). The write re-read the proofs, proved the one step the manuscript only sketches (the degree bound in the proof of Theorem 17.1), proved that the count half of the growing Poisson window is sharp (note in Section 12), and reran the delivered programs (note in Section 19). It found no false statement. Two statements are sketched or left unasserted, and are now Questions 20.7 and 20.8. *Non-claims kept from the delivery.* Finite checks do not verify the infinite quantifiers; the 80- and 120-digit computations are not interval certificates, and only the threshold certificate of Section 19 is exact; “all orders” means every fixed

order with a proved remainder, not convergence, Borel summability, uniformity in J , exponential completeness or a Stokes analysis (Remark 15.3); real-rootedness is proved for integer n only and is not used for the real-exponent inverse; the Bernoulli variables are distributional, not visible choices; Theorem 14.1 gives absolute, not large-deviation, errors; the explicit diagonal constants are inherited from Part I; interlacing and Poisson approximation are classical; no worldwide priority and no new general method are claimed; the delivered coefficient generator refuses orders above 8 as a practical guard, not a mathematical limit, and its order-4 output is an example, not a complexity claim; no OEIS entry or other external record was written. *What Part II answers in Part I.* Part I's sentence after (3.5) ("No growing-direction or boundary-uniform assertion is made") and the last sentence of its Section 6 ("No local limit theorem ... is claimed") remain true of Part I; Part II proves boundary results on the $m/n \rightarrow 0$ side (finite bounds for all m, n , the degenerate, Poisson and Gaussian regimes for every $n(m)$, a complete expansion of $A_{m,n}$ in the window $n = m(\log m + O(1))$), distributional results everywhere the variance diverges (including $m/n \rightarrow \infty$), and a local limit theorem on the diagonal. It answers Part I's Questions 2 and 4 in part: a uniform count approximation across the transition (Question 20.3), count asymptotics on the $m/n \rightarrow \infty$ side, growing marking ranges and large deviations remain open. Questions 1 and 3 of Part I are untouched; Questions 20.5 and 20.6 pose their analogues for the new expansions.

[write] Reading conventions of Part II. Part II reuses several letters of Part I with other meanings, and the manuscript reused some letters internally. The write renamed the internal clashes (second table) and fixed the readings of the rest (first table). No normalization was changed.

symbol	in Part II	not to be read as
λ	$(m/2)e^{-n/m}$, the asymptotic Poisson parameter	Part I's direction $\lambda = m/n$ of (3.1)
D	the defect $m - K$	Part I's growth constant $D = 3.16108\dots$ of (4.1), or $D(s)$ of its Section 6
r	n/m	any of Part I's four r
$\theta; \Delta$	w_1 , the exact first coefficient; $w_1^2 - 2w_2$	Part I's integration variable θ in (3.3)
$R_{m,n}, R_m(t)$	$A_{m,n}/(m!m^n) = 1/\mathbb{P}(D = 0)$ and its real-exponent continuation	Part I's tail R_T or its function $R(y)$
$d_j(m)$	defect coefficients (10.6)	Part I's d_j of (3.4)
$c_h(\lambda, r); c, c_m$	critical-window coefficient polynomials; the window parameter in $n = m(\log m + c)$, $c_m = n/m - \log m$	Part I's c_j of (3.4), or its contour abscissa c of (2.1)
$L_h(\lambda, r)$	logarithmic count coefficients (16.3)	Part I's $L = \log A$ of (5.2)
$p_i; p_h(j, r)$	Bernoulli probabilities (Sections 11, 12, 14); defect polynomials, always with arguments (Sections 15, 16)	Part I's $p(z)$, $p = x/y$, or a prime
x	$(k - M)/\sqrt{V}$ in Section 14	Section 18 uses Part I's saddle values x, y, B , with Part I's meaning
$q; \beta, \ell$	target probability; $\beta = -\log q$, $\ell = \log(m/(2\beta))$	Part I's contour variable $q = e^x - 1$
M, V	$\mathbb{E}D$ and $\text{Var } D$	Part I's index sum M of (3.4), or $M(A)$ of (5.4)
$H_a; N_q(m)$	Hermite polynomials; the integer threshold (17.10)	Part I's $H_J(v)$; its Newton iterates N_r
$C_J, C_a; c, C$	constants of Theorem 14.1 and (14.5); absolute constants local to the proofs in Sections 14 and 15	Part I's C, C_J, E_J
ε	a small exponent in the proof of Theorem 14.1; $1/m$ in Sections 15 and 17	
w_j	the weights (10.3)	Part I's Lambert value w of (5.2)

symbol	in Part II	<i>not</i> to be read as
B_{2a}	Bernoulli numbers (Section 15)	Part I's B ; Section 18's B is Part I's
$K, A_{m,n}, T_{m,k}, \mu, \sigma^2, \mu_0$	as in Part I	

Part II	manuscript	why renamed
$\psi(t) = (e^t - 1)/t$	$h(t)$	h is the expansion index throughout
$\Lambda(t) = \log \psi(t), \gamma_a$	$L(t), \ell_a$	L_h are the count coefficients; $\ell = \log(m/(2\beta))$
$\mathcal{E}_h, \mathcal{B}_h$	E_h, B_h	$E_{\pm}$ bound e ; B and B_{2a} are taken
ξ_i	B_i	Bernoulli variables, not Bernoulli numbers
$ b = \sum b_a$; product index g	$B; \ell$	B is taken; $\ell = \log(m/(2\beta))$
T_a	T_a	Touchard polynomials, not $T_{m,k} = k!S(m, k)$
$[\lambda_-, \lambda_+], O_{\lambda_{\pm}, \dots}$	$[a, b], O_{a,b, \dots}$	a, b index sums and b_a is a profile
$\mathcal{M}_h$	M_h	$M = \mathbb{E}D$
$\tau_h(\beta, \ell)$	$d_h(\beta, \ell)$	the manuscript's d_0, d_1 meant both $d_j(m)$ and the inverse corrections
η	E	the envelope error, not $E_{\pm}$ or $\mathcal{E}_h$

Label keys keep the manuscript's names (for example `a122:pd:eq:d0` for τ_0). The data files and programs keep the delivered names ($E_{\pm}$ appear as `e_lower`, `e_upper`; the inverse corrections as `d0`, `d1`).

9 Research target and the boundary left by the earlier report

OEIS A122399 begins

$$1, 1, 9, 211, 9285, 658171, 68504709, 9837380491, \dots$$

and is defined by

$$a_n = \sum_{k=0}^n k^n k! S(n, k), \quad (9.1)$$

with $0^0 = 1$ at the origin [1]. Here $S(m, k)$ denotes a Stirling number of the second kind. The ProveIt manuscript [3], located in the report directory

```
SetTheory/Cardinals/docs/reports/
generating-functions-and-asymptotics/
oeis-sequence-asymptotics/a122399-surjection-diagonal
```

proves a complete fixed-direction saddle expansion for

$$A_{m,n} = \sum_{k=1}^m k^n k! S(m, k), \quad m \geq 1, \quad n \geq 0. \quad (9.2)$$

The diagonal $A_{n,n}$ is (9.1) for $n \geq 1$. In particular, the earlier report's asymptotic remainder is uniform when m/n stays in a fixed compact subset of $(0, \infty)$. Its block-count section gives a diagonal central limit theorem but explicitly does not claim a local limit theorem or a growing-marking-range result. These are precise scope boundaries, not automatically conjectures of the wider literature.

[write] **Added 5 October 2026, batch 98.** “The ProveIt manuscript” and “the earlier report” of this Part are Part I above; [3] names its pre-layout source `proof.md`, which carries the same mathematics (see Part I's provenance note). The two scope boundaries quoted are Part I's sentence after (3.5) and the last sentence of its Section 6; both now carry dated notes pointing here.

This article fills two parts of that boundary. First, the region $n/m \sim \log m$ lies outside every such compact-direction set; it turns out to have a finite-defect expansion rather than an ordinary fixed-direction Gaussian saddle description. Second, the entire family has an exact independent-Bernoulli representation. That representation supplies local limit and Edgeworth expansions with no restriction on the ratio m/n , provided only that the variance tends to infinity.

9.1 What is inherited and what is established here

The counting interpretation, the bivariate generating function, the fixed-direction leading equivalent, and the diagonal constants in section 18 are inherited from [3]. The OEIS leading equivalent is already attributed there to Václav Kotěšovec; it is not presented here as a new conjecture. The associated growth constant has its own OEIS entry A317855 [2].

The main additional results proved here are the finite bounds in theorem 12.1, the all-direction trichotomy in theorem 13.1, the complete critical-window expansion in theorem 15.1, and the inverse-threshold expansion in theorem 17.1. The real-rootedness argument and the probabilistic tools are classical in character. Interlacing methods of this kind belong to the established framework surveyed and developed by Liu and Wang [4]; Poisson approximation for independent Bernoulli sums is classical [5]. We give the needed arguments directly. The nearby asymptotic theory of lonesum matrices [6] is relevant context, but its generating function and counting family differ from (9.2).

[write] Added 5 October 2026, batch 98: two missing credits. The manuscript credits its real-rootedness argument only generically. Two classical sources should be named. (i) The case $n = 0$ of theorem 11.1 is equivalent to Frobenius's theorem that the Eulerian polynomials have only real, simple, negative zeros [7]: with $A_m(x) = \sum_k k! S(m, k)(x-1)^{m-k}$ the Eulerian polynomial, the polynomial of the theorem at $n = 0$ is $\mathcal{P}_{m,0}(u) = u^m A_m(1+1/u)$, and $x \mapsto u = 1/(x-1)$ maps $(-\infty, 0)$ onto $(-1, 0)$. (ii) Passing from a real-rooted generating polynomial to an exact sum of independent Bernoulli variables, and from there to central and local limit theorems, is Harper's method [8], introduced for the Stirling numbers $S(m, k)$ themselves. corollary 11.2 and theorem 14.1 apply it to this array; no novelty is claimed for the method. Both items were added by the write; the write did not consult the two originals.

A targeted repository and literature inspection did not identify these specific boundary formulas in the sources examined. Such an inspection cannot establish an exhaustive priority claim. The appropriate conclusion is that the article supplies explicit, proved extensions of the inspected Provelt manuscript, with broader novelty left for independent assessment.

9.2 A guide to the principal formulas

Let $K = K_{m,n}$ be the number of blocks in a uniformly chosen structure, and let $D = m - K$ be its defect. Define

$$r = \frac{n}{m}, \quad \lambda = \frac{m}{2} e^{-r}, \quad R_{m,n} = \frac{A_{m,n}}{m! m^n}, \quad \theta = \frac{m-1}{2} \left(1 - \frac{1}{m}\right)^n. \quad (9.3)$$

The parameter θ is an exact coefficient; λ is the more convenient asymptotic coordinate. They must not be confused. For $n = m(\log m + c) + o(m)$,

$$D \implies \text{Poisson}(e^{-c}/2), \quad R_{m,n} \longrightarrow \exp(e^{-c}/2). \quad (9.4)$$

The first correction, with the actual $r = n/m$ and λ of (9.3), is

$$\log R_{m,n} = \lambda - \frac{(r/2 + 1)\lambda + (r/2 + 5/6)\lambda^2}{m} + O\left(\frac{(1+r)^2}{m^2}\right). \quad (9.5)$$

The use of the actual n retains the effect of integer rounding. The all-orders theorem below makes (9.5) part of a systematic power–logarithmic expansion.

10 Exact combinatorics and a finite-defect coefficient identity

A structure counted by $A_{m,n}$ consists of an ordered partition of a red m -element set into k nonempty blocks, together with an arbitrary map from a blue n -element set to those blocks. There are $k!S(m, k)$ ordered partitions and k^n maps. Thus

$$\mathbb{P}(K = k) = \frac{k^n k! S(m, k)}{A_{m,n}}. \quad (10.1)$$

With u marking blocks, the bivariate exponential generating function is

$$\sum_{m,n \geq 0} A_{m,n}(u) \frac{x^m}{m!} \frac{y^n}{n!} = \frac{1}{1 - ue^y(e^x - 1)}. \quad (10.2)$$

Here the extension at $m = 0$ is $A_{0,0} = 1$, $A_{0,n} = 0$ for $n > 0$. Indeed, one block contains a nonempty red set and an arbitrary blue set, so it contributes $u(e^x - 1)e^y$; an ordered sequence of blocks gives the geometric series in (10.2).

Write

$$w_j = \frac{(m-j)! S(m, m-j)}{m!} \left(1 - \frac{j}{m}\right)^n, \quad \mathcal{Q}_{m,n}(z) = \sum_{j=0}^{m-1} w_j z^j. \quad (10.3)$$

Then

$$w_0 = 1, \quad \mathcal{Q}_{m,n}(1) = R_{m,n}, \quad \mathbb{E}z^D = \frac{\mathcal{Q}_{m,n}(z)}{\mathcal{Q}_{m,n}(1)}. \quad (10.4)$$

The denominator $m!m^n$ counts the structures whose red partition is all singletons. Thus $\mathbb{P}(D = 0) = R_{m,n}^{-1}$.

Lemma 10.1 (Exact defect identity). *Set*

$$\psi(t) = \frac{e^t - 1}{t} = \sum_{a \geq 0} \frac{t^a}{(a+1)!}. \quad (10.5)$$

For $0 \leq j \leq m-1$,

$$d_j(m) := \frac{(m-j)! S(m, m-j)}{m!} = [t^j] \psi(t)^{m-j}. \quad (10.6)$$

For fixed j , $d_j(m)$ is a polynomial in m of degree j , with leading coefficient $1/(2^j j!)$. In addition,

$$0 \leq w_j \leq \frac{\lambda^j}{j!}, \quad 1 \leq R_{m,n} \leq e^\lambda. \quad (10.7)$$

These identities and inequalities also hold when the exponent n is replaced by any real $t \geq 0$.

Proof. The exponential generating function for $S(m, k)$ gives $k!S(m, k)/m! = [t^m](e^t - 1)^k$. Put $k = m - j$ and factor out t^{m-j} . This proves (10.6). Expanding the power of ψ gives the finite profile formula

$$d_j(m) = \sum_{\substack{b_1, b_2, \dots \geq 0 \\ \sum ab_a = j}} (m-j) \prod_{a \geq 1} \frac{1}{b_a! ((a+1)!)^{b_a}}, \quad |b| = \sum_{a \geq 1} b_a. \quad (10.8)$$

Only finitely many b_a are nonzero. The term $b_1 = j$ has degree j ; all other terms have lower degree. This proves the polynomial assertion.

Since $(a+1)! \geq 2^a$, coefficientwise comparison of power series with nonnegative coefficients gives

$$\psi(t) \preceq (1 - t/2)^{-1}, \quad d_j(m) \leq 2^{-j} \binom{m-1}{j} \leq \frac{m^j}{2^j j!}.$$

Also $(1 - j/m)^n \leq e^{-nj/m}$ for real $n \geq 0$. Combining the two estimates proves the first part of (10.7); summing proves the second. $\square$

Remark 10.2. The coefficientwise domination in (10.7) is useful for summation of asymptotic series: it controls every defect, including those for which a fixed- j expansion is inappropriate. It does not by itself imply a Poisson approximation, because the normalization is $R_{m,n}$, not e^λ .

11 Real zeros and an exact independent-Bernoulli representation

Define the block polynomial

$$\mathcal{P}_{m,n}(u) = \sum_{k=1}^m k^n k! S(m, k) u^k. \quad (11.1)$$

This section assumes that n is a nonnegative integer. The real-exponent continuation used later for inversion will not be assumed to preserve real-rootedness.

Theorem 11.1 (Root location and interlacing). *For $m \geq 1$ and integer $n \geq 0$, $\mathcal{P}_{m,n}$ has a simple zero at zero. Its other $m-1$ zeros are simple and lie in $(-1, 0)$. Consequently there exist distinct numbers $\alpha_1, \dots, \alpha_{m-1} \in (0, 1)$ such that*

$$\mathcal{P}_{m,n}(u) = m! m^n u \prod_{i=1}^{m-1} (u + \alpha_i), \quad \mathcal{Q}_{m,n}(z) = \prod_{i=1}^{m-1} (1 + \alpha_i z). \quad (11.2)$$

As n increases by one, the nonzero zeros strictly interlace and move towards zero in their ordered positions.

Proof. Let $F_m(u) = \mathcal{P}_{m,0}(u)$. The Stirling recurrence gives

$$F_{m+1}(u) = u((1+u)F_m(u))'. \quad (11.3)$$

The induction starts with $F_1(u) = u$. If F_m has its simple zeros in $(-1, 0]$, with zero as its rightmost zero, then $(1+u)F_m$ has $m+1$ simple zeros in $[-1, 0]$. Rolle's theorem places one zero of its derivative in each of the m intervening open intervals. The derivative has degree m , so these are all its zeros and all are simple. Multiplication by u adds one simple zero at zero. This proves the claim for $n = 0$.

For general integer n ,

$$\mathcal{P}_{m,n+1}(u) = u\mathcal{P}'_{m,n}(u). \quad (11.4)$$

Apply the same Rolle argument to the m simple zeros of $\mathcal{P}_{m,n}$ in $(-1, 0]$. The derivative has exactly one zero between successive zeros. Multiplication by u restores the zero at zero. This also proves the asserted direction of interlacing. The leading coefficient of $\mathcal{P}_{m,n}$ is $m!m^n$, which yields the first factorization in (11.2). Reversing coefficients yields the second. $\square$

Corollary 11.2 (Bernoulli factorization). *There are independent variables $\xi_1, \dots, \xi_{m-1}$ with*

$$\xi_i \sim \text{Bernoulli}(p_i), \quad p_i = \frac{\alpha_i}{1 + \alpha_i} \in (0, 1/2), \quad (11.5)$$

such that

$$D \stackrel{\text{law}}{=} \sum_{i=1}^{m-1} \xi_i. \quad (11.6)$$

Equivalently, K is one plus a sum of $m-1$ independent Bernoulli variables with success probabilities in $(1/2, 1)$.

Proof. Divide the second product in (11.2) by its value at $z = 1$:

$$\mathbb{E}z^D = \prod_i \frac{1 + \alpha_i z}{1 + \alpha_i} = \prod_i (1 - p_i + p_i z).$$

This is the probability generating function of the stated independent sum. $\square$

The independent variables in (11.6) are a distributional representation; they are not claimed to be directly visible independent choices in the original ordered partition. Nevertheless, the representation is exact and immediately gives concentration, log-concavity, and limit theorems. For example, the coefficient sequence of $\mathcal{Q}$ is ultra-log-concave: dividing w_j by $\binom{m-1}{j}$ gives a log-concave sequence. This follows from Newton's inequalities for the elementary symmetric functions of the positive α_i . The main use below is quantitative rather than qualitative.

[write] **Added 5 October 2026, batch 98: formal status.** The identity $\sum_{k \leq m} k! S(m, k) (X - 1)^{m-k} = A_m(X)$ used in the note of section 9 is formalized in Lean as `Fabius.permutohedron_h_polynomial` (`Analysis/FabiusFunction/Lean/FabiusFunction/EulerianPermutohedron.lean`). No real-rootedness, interlacing or Bernoulli factorization is formalized anywhere in the repository; theorem 11.1 and corollary 11.2 and everything after them are unformalized (question 20.6 asks for that layer).

12 Two coefficients give global finite-input bounds

Let

$$\theta = w_1, \quad \Delta = w_1^2 - 2w_2, \quad (12.1)$$

where $w_2 = 0$ if $m = 2$. The case $m = 1$ is deterministic and will be omitted in statements involving these parameters. Directly from the partitions with one or two defects,

$$\theta = \frac{m-1}{2} \left(1 - \frac{1}{m}\right)^n, \quad (12.2)$$

$$w_2 = \frac{(m-2)(3m-5)}{24} \left(1 - \frac{2}{m}\right)^n. \quad (12.3)$$

For (12.3) at $m = 2, n = 0$, use $0^0 = 1$; the zero prefactor still makes $w_2 = 0$.

Theorem 12.1 (Finite enumeration and approximation bounds). *For integers $m \geq 2, n \geq 0$, put $M = \mathbb{E}D$ and $V = \text{Var}(D)$. Then*

$$0 \leq \Delta \leq \theta^2 \min \left\{ 1, \frac{n + (5m-7)/3}{(m-1)^2} \right\}, \quad (12.4)$$

$$\theta - \frac{\Delta}{2} \leq \log R_{m,n} \leq \theta, \quad (12.5)$$

$$\max\{\theta/2, \theta - \Delta\} \leq M \leq \theta, \quad (12.6)$$

$$\max\{\theta/4, \theta - 2\Delta\} \leq V \leq \theta, \quad (12.7)$$

$$d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(\theta)) \leq \min\{1, 2\Delta\}. \quad (12.8)$$

Here $d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_j |\mu\{j\} - \nu\{j\}|$. All constants in these inequalities are explicit; no asymptotic threshold is involved.

Proof. The factorization gives

$$\theta = \sum_i \alpha_i, \quad \Delta = \sum_i \alpha_i^2. \quad (12.9)$$

Thus $\Delta \geq 0$, and (12.2)–(12.3) give the exact identity

$$\frac{\Delta}{\theta^2} = 1 - \left(1 - \frac{5m-7}{3(m-1)^2}\right) \left(1 - \frac{1}{(m-1)^2}\right)^n. \quad (12.10)$$

The two quantities subtracted from one lie in $[0, 1]$. The elementary inequality $1 - (1-a)(1-b)^n \leq a + nb$ proves (12.4); the upper bound by one is immediate.

For $x \geq 0$, $x - x^2/2 \leq \log(1+x) \leq x$. Summing this inequality over $x = \alpha_i$ proves (12.5). Also

$$M = \sum_i \frac{\alpha_i}{1 + \alpha_i}, \quad V = \sum_i \frac{\alpha_i}{(1 + \alpha_i)^2}.$$

The restrictions $0 < \alpha_i < 1$ imply $\theta/2 \leq M \leq \theta$ and $\theta/4 \leq V \leq \theta$. Moreover,

$$\theta - M = \sum_i \frac{\alpha_i^2}{1 + \alpha_i} \leq \Delta,$$

and

$$\theta - V = \sum_i \frac{\alpha_i^2(2 + \alpha_i)}{(1 + \alpha_i)^2} \leq 2\Delta.$$

This proves the moment bounds.

For completeness, the required elementary Poisson approximation is

$$d_{\text{TV}} \left(\mathcal{L} \left(\sum_i \text{Bernoulli}(p_i) \right), \text{Poisson} \left(\sum_i p_i \right) \right) \leq \sum_i p_i^2. \quad (12.11)$$

For one variable the distance is $p(1 - e^{-p}) \leq p^2$. Replace factors one at a time in the convolution of their laws; convolution with a probability law cannot increase total variation. This proves (12.11), the usual elementary Le Cam bound [5]. Since $\sum_i p_i^2 \leq \Delta$ and $0 \leq \theta - M \leq \Delta$, couple a $\text{Poisson}(M)$ variable with an independent $\text{Poisson}(\theta - M)$ increment. The resulting Poisson laws have distance at most $\theta - M$. The triangle inequality gives (12.8). $\square$

12.1 Replacing the exact parameter by the asymptotic one

For all $m \geq 2$ and $n \geq 0$,

$$0 \leq \lambda - \theta \leq \lambda \left(\frac{1}{m} + \frac{n}{2m(m-1)} \right). \quad (12.12)$$

Indeed,

$$\frac{\theta}{\lambda} = (1 - 1/m) \exp \left[-n \sum_{a \geq 2} \frac{1}{am^a} \right], \quad \sum_{a \geq 2} \frac{1}{am^a} \leq \frac{1}{2m(m-1)}.$$

Use $1 - e^{-x} \leq x$. It follows that

$$d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(\lambda)) \leq 2\Delta + \lambda - \theta. \quad (12.13)$$

In particular, if $n = O(m \log m)$,

$$d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(\lambda)) = O\left(\frac{(\lambda + \lambda^2)(1 + r)}{m}\right). \quad (12.14)$$

This is useful even when λ grows. If $\lambda^2(1 + r)/m \rightarrow 0$, with $n = O(m \log m)$ and λ bounded below by a positive constant, then

$$R_{m,n} = e^\lambda(1 + o(1)), \quad d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(\lambda)) = o(1). \quad (12.15)$$

The count statement follows from $|\log R_{m,n} - \lambda| \leq \lambda - \theta + \Delta/2$. Thus there is an explicit overlap region $1 \ll \lambda \ll \sqrt{m/\log m}$ in which a growing Poisson approximation and a Gaussian approximation are simultaneously valid. No assertion that this sufficient window is sharp is made.

[write] Added 5 October 2026, batch 98: the count half of the window is sharp. The write proves that the condition $\lambda^2(1 + r)/m \rightarrow 0$ is also *necessary* for the count statement in (12.15); the total-variation half is not settled and stays open (question 20.7).

Claim. Let $m \rightarrow \infty$ and let $n = n(m) \geq 0$ be integers with $\lambda \geq \lambda_0$ for a fixed $\lambda_0 > 0$. Then $R_{m,n}e^{-\lambda} \rightarrow 1$ if and only if $\lambda^2(1 + r)/m \rightarrow 0$.

Proof. The condition $\lambda \geq \lambda_0$ gives $r \leq \log(m/(2\lambda_0))$, so $n = O(m \log m)$ and sufficiency is (12.15). For necessity, $f(x) = x - x^2/2 + x^3/3 - \log(1 + x)$ has $f(0) = 0$ and $f'(x) = x^3/(1 + x) \geq 0$, so $\log(1 + x) \leq x - x^2/2 + x^3/3$ for $x \geq 0$. Sum over the $\alpha_i \in (0, 1)$ of (11.2), using (12.9) and $\sum_i \alpha_i^3 \leq \sum_i \alpha_i^2 = \Delta$: $\log R_{m,n} \leq \theta - \Delta/2 + \Delta/3 = \theta - \Delta/6$. With $\theta \leq \lambda$ from (12.12),

$$\log R_{m,n} - \lambda \leq -\Delta/6.$$

Next bound Δ below by (12.10). Put $a_m = (5m - 7)/(3(m - 1)^2)$ and $b_m = 1/(m - 1)^2$, both in $[0, 1]$ for $m \geq 2$. Then $(1 - a_m)(1 - b_m)^n \leq e^{-(a_m + nb_m)}$, and concavity of $1 - e^{-z}$ gives $1 - e^{-z} \geq (1 - e^{-1}) \min\{1, z\}$ for $z \geq 0$. Moreover $a_m \geq 1/m$, because $(5m - 7)m - 3(m - 1)^2 = (2m - 3)(m + 1) \geq 0$, and $nb_m \geq n/m^2 = r/m$. Hence, for all $m \geq 2$ and $n \geq 0$,

$$\Delta \geq (1 - e^{-1}) \theta^2 \min\left\{1, \frac{1 + r}{m}\right\}.$$

Finally (12.12) gives $\theta/\lambda \geq 1 - 1/m - r/(2(m - 1)) \rightarrow 1$, since $r = O(\log m)$; so $\theta \geq \lambda/2$ for all large m . If $R_{m,n}e^{-\lambda} \rightarrow 1$, then $\Delta \rightarrow 0$, hence $\lambda^2 \min\{1, (1 + r)/m\} \rightarrow 0$; as $\lambda \geq \lambda_0$, the minimum tends to zero, so it eventually equals $(1 + r)/m$ and $\lambda^2(1 + r)/m \rightarrow 0$. $\square$

In particular, for every path with $\lambda \geq \lambda_0$ (where $r = \log(m/(2\lambda)) \leq \log m + O(1)$), the count statement holds exactly when $\lambda = o(\sqrt{m/\log m})$, the upper end of the overlap region above. (Dated note, 5 October 2026: the first version of this sentence read “along $r = \log m + O(1)$ with $\lambda \rightarrow \infty$ ”, a regime that cannot occur, since $r = \log m + O(1)$ forces $\lambda = \Theta(1)$. An independent adversarial check made by the intake after the write proved the corrected statement in both directions, found the necessity proof, the degree bound $\deg_\ell \tau_h \leq h + 1$ (attained for $h \leq 5$) and τ_2 valid, and tested them against exact counts; a careful reading, not a formal verification.)

13 An all-direction Gaussian–Poisson–degenerate trichotomy

We now let $m \rightarrow \infty$ through integers and allow any nonnegative integer function $n = n(m)$. The only parameter that matters for the qualitative regime is the exact first coefficient θ .

Theorem 13.1 (Global trichotomy). *With $D = D_{m,n(m)}$, the following statements hold.*

- (i) *If $\theta \rightarrow 0$, then $D \rightarrow 0$ in total variation, with $\mathbb{P}(D \neq 0) \leq \theta$, and $R_{m,n} \rightarrow 1$.*
- (ii) *If $\theta \rightarrow \theta_0 \in (0, \infty)$, then $D \rightarrow \text{Poisson}(\theta_0)$ in total variation and $R_{m,n} \rightarrow e^{\theta_0}$.*
- (iii) *If $\theta \rightarrow \infty$, then $V \rightarrow \infty$ and $(D - M)/\sqrt{V} \Rightarrow N(0, 1)$. In fact, the uniform local and all-order Edgeworth assertions of theorem 14.1 hold.*

Equivalently, put $c_m = n/m - \log m$. If $c_m \rightarrow +\infty$ the first case applies; if $c_m \rightarrow c \in \mathbb{R}$ the second case has $\theta_0 = e^{-c}/2$; if $c_m \rightarrow -\infty$ the third case applies.

Proof. For (i), Markov's inequality gives $\mathbb{P}(D \neq 0) \leq M \leq \theta$, while $0 \leq \log R_{m,n} \leq \theta$. If θ tends to a finite nonzero limit, the exact expression (12.2) forces $n = m \log m + O(m)$: solve its logarithm for n , using $-\log(1 - 1/m) = m^{-1} + O(m^{-2})$. Consequently (12.4) gives $\Delta = O(\log m/m) \rightarrow 0$. Equations (12.8) and (12.5) prove (ii). For (iii), $V \geq \theta/4 \rightarrow \infty$; the independent-Bernoulli local theorem proved in the next section gives the conclusion.

For the equivalent description, $\lambda = e^{-c_m}/2$. If $c_m \rightarrow +\infty$, then $\theta \leq \lambda \rightarrow 0$. In the finite- c_m case, (12.12) gives $\theta/\lambda \rightarrow 1$. If $c_m \rightarrow -\infty$, then eventually $n \leq m \log m$; again (12.12) gives $\theta/\lambda \rightarrow 1$, and $\lambda \rightarrow \infty$. $\square$

Remark 13.2 (Centering in the dense Gaussian region). The Gaussian statement is centered at the exact mean M and scaled by the exact variance V . Replacing both by θ without a further error estimate would be incorrect in general. For example, the ordinary diagonal $n = m$ lies well inside the Gaussian region but not inside the small- Δ window. The bounds in theorem 12.1 deliberately separate these situations.

14 Local limit laws and all-order Edgeworth expansions

This section proves the local theorem needed above without assuming a fixed direction or a separate saddle-point representation. Let κ_a denote the cumulants of D , so that

$$\kappa_1 = M, \quad \kappa_2 = V, \quad \kappa_3 = \sum_i p_i(1 - p_i)(1 - 2p_i).$$

Write

$$\varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad H_a(x) = (-1)^a e^{x^2/2} \frac{d^a}{dx^a} e^{-x^2/2}$$

for the Gaussian density and the probabilists' Hermite polynomials.

Theorem 14.1 (Uniform lattice Edgeworth expansion). *For every fixed integer $J \geq 0$, there is a constant C_J independent of m, n such that, as $V \rightarrow \infty$,*

$$\mathbb{P}(D = k) = \frac{\varphi(x)}{\sqrt{V}} \sum_{\substack{b_3, \dots, b_{J+2} \geq 0 \\ \sum_{a=3}^{J+2} (a-2)b_a \leq J}} H_{\sum a b_a}(x) \prod_{a=3}^{J+2} \frac{1}{b_a!} \left(\frac{\kappa_a}{a! V^{a/2}} \right)^{b_a} + O_J(V^{-(J+2)/2}), \quad (14.1)$$

uniformly for all integers k , where $x = (k - M)/\sqrt{V}$. The empty multi-index contributes 1. In particular,

$$\sup_{k \in \mathbb{Z}} \left| \sqrt{V} \mathbb{P}(D = k) - \varphi\left(\frac{k - M}{\sqrt{V}}\right) \right| = O(V^{-1/2}), \quad (14.2)$$

$$\mathbb{P}(D = k) = \frac{\varphi(x)}{\sqrt{V}} \left[1 + \frac{\kappa_3}{6V^{3/2}}(x^3 - 3x) \right] + O(V^{-3/2}). \quad (14.3)$$

The same result applies to $K = m - D$ by reflection.

Proof. We record the two ingredients that make the constants uniform across this entire triangular family. First, for real t ,

$$|1 - p + pe^{it}|^2 = 1 - 4p(1 - p) \sin^2(t/2).$$

Using $1 - x \leq e^{-x}$ and multiplying the independent factors gives

$$\left| \mathbb{E} e^{it(D-M)} \right| \leq \exp\{-2V \sin^2(t/2)\}, \quad |t| \leq \pi. \quad (14.4)$$

There is no competing lattice peak in $(-\pi, \pi)$.

Second, on a fixed small complex disk about zero,

$$\log(1 + p(e^z - 1)) - pz$$

is analytic uniformly for $0 \leq p \leq 1/2$. After division by $p(1 - p)$ it has a removable limit at $p = 0$ and is uniformly bounded on a still smaller disk. Cauchy's estimate therefore implies, for every fixed $a \geq 2$,

$$|\kappa_a| \leq C_a V. \quad (14.5)$$

It also bounds the Taylor remainder of the cumulant series by $C_A V |z|^{A+1}$ through any fixed order A .

Use lattice Fourier inversion and put $t = s/\sqrt{V}$:

$$\sqrt{V} \mathbb{P}(D = k) = \frac{1}{2\pi} \int_{-\pi\sqrt{V}}^{\pi\sqrt{V}} e^{-isx} \mathbb{E} e^{is(D-M)/\sqrt{V}} ds. \quad (14.6)$$

For $|s| \leq V^\varepsilon$, with $\varepsilon > 0$ sufficiently small for the fixed order J , expand

$$\mathbb{E} e^{is(D-M)/\sqrt{V}} = e^{-s^2/2} \exp\left\{ \sum_{a \geq 3} \frac{\kappa_a (is)^a}{a! V^{a/2}} \right\}.$$

By (14.5), the term indexed by a multi-index (b_a) has order at most $V^{-\frac{1}{2} \sum (a-2)b_a}$ times a fixed polynomial in $|s|$. Retain exactly the indices of total grade at most J . Taylor's theorem, the uniform analytic remainder bound, and a slight weakening of the Gaussian exponent bound the integrated discarded part by $O_J(V^{-(J+1)/2})$. More explicitly, the local remainder is bounded by that power of V times a fixed polynomial in $|s|$ times e^{-cs^2} for an absolute $c > 0$, after shrinking ε if necessary.

For $V^\varepsilon < |s| \leq \delta\sqrt{V}$, a fixed sufficiently small δ and (14.4) give a bound e^{-cs^2} ; its integral is smaller than every inverse power of V . For $\delta\sqrt{V} \leq |s| \leq \pi\sqrt{V}$, (14.4) gives an exponentially small integral. Thus the retained polynomial–Gaussian integrals may be extended to the whole real line with the same error.

Finally,

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-isx} e^{-s^2/2} (is)^a ds = H_a(x) \varphi(x).$$

Termwise integration gives (14.1); division by $\sqrt{V}$ produces its stated error. All estimates before integration use $|e^{-isx}| = 1$, so the error is uniform in k . $\square$

Remark 14.2. The error in (14.1) is an absolute error. It is not a relative large-deviation estimate arbitrarily far into the tails. Also, using exact cumulants is important: the theorem gives a uniform expansion for any diverging-variance path $n(m)$ without first requiring asymptotic formulas for those cumulants.

15 The critical-window expansion to every fixed order

The limit theorem (9.4) does not quantify the logarithmic finite-size corrections. We now construct them, including a uniform remainder that justifies summing over all defects. Throughout this section $r = n/m \geq 0$ and $\lambda = (m/2)e^{-r}$, but n may be a nonnegative real number.

15.1 A finite rational coefficient algorithm

Use the convergent germ

$$\Lambda(t) = \log \psi(t) = \sum_{a \geq 1} \gamma_a t^a = \frac{t}{2} + \frac{t^2}{24} - \frac{t^4}{2880} + \frac{t^6}{181440} - \cdots. \quad (15.1)$$

Thus $\gamma_1 = 1/2$, $\gamma_{2a} = B_{2a}/((2a)(2a)!)$, and $\gamma_{2a+1} = 0$ for $a \geq 1$. These identities may also be obtained by formal division and logarithm of (10.5); no knowledge of a Bernoulli-number convention is needed by the construction.

For indeterminates v, j, r , set

$$\mathcal{E}_h(v, j, r) = 2^{h+1} \gamma_{h+1} v^{h+1} - j 2^h \gamma_h v^h - \frac{r j^{h+1}}{h+1}, \quad h \geq 1. \quad (15.2)$$

Define polynomials $\mathcal{B}_h(v, j, r)$ by

$$\mathcal{B}_0 = 1, \quad \mathcal{B}_h = \frac{1}{h} \sum_{a=1}^h a \mathcal{E}_a \mathcal{B}_{h-a}. \quad (15.3)$$

Equivalently, $\mathcal{B}_h$ is the coefficient of ε^h in $\exp(\sum_{a \geq 1} \mathcal{E}_a \varepsilon^a)$. Let $\mathcal{T}_j$ be the linear map acting on powers of v by

$$\mathcal{T}_j(v^a) = (j)_{\underline{a}}, \quad (15.4)$$

leaving coefficients in j, r untouched, and define

$$p_h(j, r) = \mathcal{T}_j \mathcal{B}_h(v, j, r). \quad (15.5)$$

Finally, define the Poisson transform

$$c_h(\lambda, r) = e^{-\lambda} \sum_{j \geq 0} \frac{\lambda^j}{j!} p_h(j, r). \quad (15.6)$$

This is a polynomial, not an infinite object that must be evaluated by summation: replace each monomial j^a in p_h by the Touchard polynomial

$$\mathsf{T}_a(\lambda) = \sum_{b=0}^a S(a, b) \lambda^b. \quad (15.7)$$

One has

$$\deg_j p_h \leq 2h, \quad \deg_r p_h \leq h, \quad \deg_\lambda c_h \leq 2h, \quad \deg_r c_h \leq h. \quad (15.8)$$

All coefficients are rational, and $p_0 = c_0 = 1$.

Theorem 15.1 (Complete boundary expansion). *Fix $0 < \lambda_- < \lambda_+ < \infty$, $Z > 0$, and an integer $J \geq 0$. Uniformly for real $n \geq 0$ with $\lambda \in [\lambda_-, \lambda_+]$ and for complex z with $|z| \leq Z$,*

$$\mathcal{Q}_{m,n}(z) = e^{\lambda z} \left\{ \sum_{h=0}^J \frac{c_h(\lambda z, r)}{m^h} + O_{\lambda_\pm, Z, J} \left(\frac{(1+r)^{J+1}}{m^{J+1}} \right) \right\}. \quad (15.9)$$

In particular,

$$A_{m,n} = m! m^n e^\lambda \left\{ \sum_{h=0}^J \frac{c_h(\lambda, r)}{m^h} + O_{\lambda_\pm, J} \left(\frac{(1+r)^{J+1}}{m^{J+1}} \right) \right\}. \quad (15.10)$$

Here $r = \log(m/(2\lambda)) = \log m + O(1)$. The expansion can be differentiated any fixed number of times with respect to z , with uniform remainders on smaller fixed disks.

15.2 Formal derivation of the coefficient algorithm

Put $\varepsilon = 1/m$ and replace t by $2\varepsilon v$ in (10.6). Then

$$G_j(\varepsilon) := j!(2\varepsilon)^j d_j(1/\varepsilon) = j![v^j] \exp\{(\varepsilon^{-1} - j)\Lambda(2\varepsilon v)\}. \quad (15.11)$$

The apparent singularity at $\varepsilon = 0$ is removable. Expanding the exponent gives

$$(\varepsilon^{-1} - j)\Lambda(2\varepsilon v) = v + \sum_{h \geq 1} (2^{h+1}\gamma_{h+1}v^{h+1} - j2^h\gamma_h v^h)\varepsilon^h.$$

The exponential tilt contributes

$$(1 - j\varepsilon)^{r/\varepsilon} e^{rj} = \exp\left\{-r \sum_{h \geq 1} \frac{j^{h+1}\varepsilon^h}{h+1}\right\}. \quad (15.12)$$

For any nonnegative integer j , $j![v^j]e^v v^a = (j)_{\underline{a}}$. Combining (15.11) and (15.12) yields

$$w_j \sim \frac{\lambda^j}{j!} \sum_{h \geq 0} \frac{p_h(j, r)}{m^h}. \quad (15.13)$$

The degree bounds follow directly from (15.2): a term of grade h has degree at most $2h$ after applying $\mathcal{T}_j$, and at most h factors of r . Summing (15.13) formally gives (15.6). The next subsection supplies the missing uniform justification.

15.3 Uniform control of the defect expansion

Lemma 15.2 (A polynomial Cauchy bound). *There are absolute positive constants c, C such that, for every integer $j \geq 0$ and real $r \geq 0$, the function*

$$F_{j,r}(\varepsilon) = G_j(\varepsilon) \exp\left\{-r \sum_{h \geq 1} \frac{j^{h+1}\varepsilon^h}{h+1}\right\} \quad (15.14)$$

is analytic and bounded in modulus by C on

$$|\varepsilon| \leq \rho_{j,r} := \frac{c}{(1+r)(1+j)^2}. \quad (15.15)$$

Its Taylor coefficient of order h is $p_h(j, r)$. For $0 \leq \varepsilon \leq \rho_{j,r}/2$,

$$\left|F_{j,r}(\varepsilon) - \sum_{h=0}^J p_h(j, r)\varepsilon^h\right| \leq C_J((1+r)(1+j)^2\varepsilon)^{J+1}. \quad (15.16)$$

Proof. The coefficient statement follows from the formal derivation, so it suffices to obtain the analytic bound. For $j = 0$ the function is one. For $j \geq 1$, multiply (10.8) by $j!(2\varepsilon)^j$ and put $m = \varepsilon^{-1}$. A profile with $|b| = \sum b_a$ contributes

$$\frac{j!}{\prod b_a!} \prod_{a \geq 1} \left(\frac{2^a}{(a+1)!}\right)^{b_a} \varepsilon^{j-|b|} \prod_{g=0}^{|b|-1} (1 - (j+g)\varepsilon).$$

The last product has modulus at most $\exp(2j^2|\varepsilon|)$. After fixing b_a for $a \geq 2$, the value $b_1 = j - \sum_{a \geq 2} ab_a$ is forced. Use $j!/b_1! \leq j^{\sum_{a \geq 2} ab_a}$ and then enlarge the sum by allowing all choices of b_a for $a \geq 2$. This gives

$$|G_j(\varepsilon)| \leq \exp \left\{ 2j^2|\varepsilon| + \sum_{a \geq 2} \frac{2^a}{(a+1)!} j^a |\varepsilon|^{a-1} \right\}. \quad (15.17)$$

When $j|\varepsilon| \leq 1/4$, the sum in this exponent is bounded by a constant times $j^2|\varepsilon|$. The series in the second exponential of (15.14) is analytic for $|j\varepsilon| < 1$, and its modulus is bounded by $Crj^2|\varepsilon|$ on the same disk. Choosing c sufficiently small in (15.15) bounds both exponents by absolute constants. This proves the uniform analytic bound. Cauchy's coefficient estimate and summation of a geometric tail at $|\varepsilon|/\rho_{j,r} \leq 1/2$ give (15.16). $\square$

Proof of theorem 15.1. Because $\lambda \in [\lambda_-, \lambda_+]$, one has $r = \log m + O_{\lambda_{\pm}}(1)$. Split the defect sum at $j_* = \lfloor m^{1/4} \rfloor$. For $j \leq j_*$, the condition in lemma 15.2 holds for all sufficiently large m , uniformly in the parameter range. Multiply (15.16) by $|\lambda z|^j/j!$ and sum. Since

$$\sum_{j \geq 0} \frac{(\lambda_+ Z)^j}{j!} (1+j)^{2J+2} < \infty,$$

the total remainder is $O_{\lambda_{\pm}, Z, J}((1+r)^{J+1}m^{-J-1})$.

For $j > j_*$, (10.7) bounds the original terms by $(\lambda_+ Z)^j/j!$. This tail is smaller than every fixed inverse power of m . The tails of the approximating terms have the same property: by (15.8) they are bounded by a fixed power of j and of $1+r$ times $(\lambda_+ Z)^j/j!$. Thus each truncated sum may be extended to all $j \geq 0$. Formula (15.6) then proves (15.9). Factoring out $e^{\lambda z}$ does not change the order of the uniform error, since its reciprocal is bounded on the fixed disk. Setting $z = 1$ proves (15.10). Finally, apply Cauchy's derivative estimate to a slightly larger disk to obtain the differentiated statements. $\square$

Remark 15.3 (Meaning of “all orders”). For every fixed J there is a proved remainder of the indicated order. The article does not assert convergence of the infinite series in h , a uniform bound as $J \rightarrow \infty$, Borel summability, or an exponentially improved remainder. After substituting $r = \log m - \log(2\lambda)$, the result is a complete power–logarithmic, or perturbative transseries, expansion. Exponentially small sectors have not been identified here.

[write] Added 5 October 2026, batch 98: what “transseries” means here. The manuscript's subtitle (“Boundary transseries, local limit laws, and inverse thresholds”) and the name of theorem 17.1 (“Inverse critical transseries”) use the word for what this remark describes exactly: a power–logarithmic expansion in m^{-h} whose coefficients are polynomials in $r = \log m - \log(2\lambda)$ (in theorem 17.1, in ℓ), proved to every fixed order. It has logarithmic depth one and no exponentially small terms, and nothing transseries-specific (transmonomials beyond $m^{-h}(\log m)^k$, Borel resummation, resurgence, Stokes data) is claimed or used. The word overstates what is proved; read it as “power–logarithmic expansion” throughout this Part. The delivered wording is kept on record; for the same reason the report is placed in this collection and not under **Analysis/Transseries**.

16 Explicit corrections and the geometry of rare defects

The first two defect polynomials are

$$p_1(j, r) = -\frac{(3r+5)j^2 + j}{6}, \quad (16.1)$$

$$p_2(j, r) = \frac{j}{72} \left[(9r^2 + 30r + 25)j^3 - (18r + 18)j^2 - j - 6 \right]. \quad (16.2)$$

It is shorter to print the logarithmic count coefficients. Define $L_h(\lambda, r)$ by

$$L_1 = c_1, \quad L_h = c_h - \frac{1}{h} \sum_{a=1}^{h-1} a L_a c_{h-a} \quad (h \geq 2). \quad (16.3)$$

Then

$$\log R_{m,n} = \lambda + \sum_{h=1}^J \frac{L_h(\lambda, r)}{m^h} + O\left(\frac{(1+r)^{J+1}}{m^{J+1}}\right). \quad (16.4)$$

The first two are

$$L_1 = -\left(\frac{r}{2} + 1\right)\lambda - \left(\frac{r}{2} + \frac{5}{6}\right)\lambda^2, \quad (16.5)$$

$$L_2 = \left(\frac{r^2}{8} + \frac{r}{6}\right)\lambda + \left(\frac{3r^2}{4} + \frac{5r}{3} + \frac{7}{6}\right)\lambda^2 + \left(\frac{r^2}{2} + \frac{4r}{3} + 1\right)\lambda^3. \quad (16.6)$$

Thus $c_1 = L_1$, $c_2 = L_2 + L_1^2/2$. The third logarithmic coefficient is printed in appendix A; the package supplies exact coefficients through the fourth order and an algorithm for any prescribed finite order.

16.1 The full defect law, not just the total count

The complex-marked expansion yields, on any fixed disk and for large m , a branch of the logarithm with

$$\log \mathbb{E} z^D = \lambda(z-1) + \sum_{h=1}^J \frac{L_h(\lambda z, r) - L_h(\lambda, r)}{m^h} + O\left(\frac{(1+r)^{J+1}}{m^{J+1}}\right). \quad (16.7)$$

To see that the logarithm is legitimate, (15.9) shows uniformly that $\mathcal{Q}_{m,n}(z)e^{-\lambda z} = 1 + o(1)$ on that disk. In particular the polynomial is zero-free there for sufficiently large m . The logarithm near one is then unambiguously continued across the disk after factoring out $e^{\lambda z}$.

For every fixed defect j ,

$$\mathbb{P}(D = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 + \frac{p_1(j, r) - c_1(\lambda, r)}{m} + O_j\left(\frac{(1+r)^2}{m^2}\right) \right]. \quad (16.8)$$

This is a coefficient statement, whereas (16.7) simultaneously encodes factorial moments and cumulants. Putting $z = e^s$ and differentiating at $s = 0$ gives

$$M = \lambda - \frac{(r/2 + 1)\lambda + (r + 5/3)\lambda^2}{m} + O\left(\frac{(1+r)^2}{m^2}\right), \quad (16.9)$$

$$V = \lambda - \frac{(r/2 + 1)\lambda + (2r + 10/3)\lambda^2}{m} + O\left(\frac{(1+r)^2}{m^2}\right). \quad (16.10)$$

In particular,

$$M - V = \frac{(r + 5/3)\lambda^2}{m} + O\left(\frac{(1+r)^2}{m^2}\right) > 0$$

for large m in a compact critical window. The exact inequality $M > V$ already follows from the Bernoulli representation; the expansion quantifies the first departure from Poisson variance.

16.2 Why the limiting defects are pairs

A red partition of defect j whose blocks have size at most two has exactly j doubletons and $m - 2j$ singletons. Its contribution to $d_j(m)$ is

$$d_j^{\text{pair}}(m) = \frac{(m - j)_j}{2^j j!}, \quad (16.11)$$

with the value zero when there are too few elements. For fixed j ,

$$j! \left(\frac{2}{m}\right)^j d_j^{\text{pair}}(m) = 1 - \frac{j(3j - 1)}{2m} + O_j(m^{-2}).$$

The corresponding full coefficient at $r = 0$ is $1 - (5j^2 + j)/(6m) + O_j(m^{-2})$ by (16.1). The difference is $2j(j - 1)/(3m)$.

Proposition 16.1 (First non-pair defect). *Uniformly for λ in a compact subset of $(0, \infty)$, the probability that the red partition contains a block of size at least three is*

$$\mathbb{P}(\text{some red block has size at least 3}) = \frac{2\lambda^2}{3m} + O\left(\frac{(1+r)^2}{m^2}\right). \quad (16.12)$$

Consequently, with probability tending to one, every non-singleton red block is a doubleton; their number converges to $\text{Poisson}(\lambda_0)$ when $\lambda \rightarrow \lambda_0 \in (0, \infty)$.

Proof. Subtract the pair-only coefficients from the full coefficients, multiply by the common tilt, and use the first-order difference just computed. The same profile-product estimate used in lemma 15.2 applies to the pair-only term (it is simply a finite product). Thus the difference can be summed uniformly over j , with a remainder of the order displayed in (16.12). Its unnormalized leading sum is

$$\frac{2}{3m} \sum_{j \geq 0} \frac{\lambda^j}{j!} j(j - 1) = e^\lambda \frac{2\lambda^2}{3m}.$$

Divide by $R_{m,n} = e^\lambda(1 + O((1+r)/m))$. The second statement follows because on the pair-only event the number of doubletons equals D . $\square$

This identifies the combinatorial origin of the factor $1/2$ in the critical Poisson parameter: a single red collision is a pair, with exponential generating-function weight $1/2!$. Triples first contribute one order later.

17 Inverting the transition: a prescribed singleton probability

A useful inverse problem is not to invert the already studied diagonal count, but to ask how many blue labels are needed to make the all-singleton red partition have a prescribed probability. This turns the boundary expansion into an operational threshold formula.

For integer $m \geq 2$ and real $t \geq 0$, use the exact continuation

$$R_m(t) = \sum_{j=0}^{m-1} d_j(m)(1 - j/m)^t, \quad q_m(t) = R_m(t)^{-1}. \quad (17.1)$$

The term $j = 0$ equals one and all the others are strictly decreasing, so q_m is continuous and strictly increasing, with limit one as $t \rightarrow \infty$. For every fixed $q \in (0, 1)$, a unique solution $q_m(t_q) = q$ exists for all sufficiently large m . Indeed, $R_m(0) \geq 1 + (m-1)/2 \rightarrow \infty$.

Theorem 17.1 (Inverse critical transseries). *Fix q in a compact subinterval of $(0, 1)$ and put*

$$\beta = -\log q, \quad \ell = \log \frac{m}{2\beta}. \quad (17.2)$$

Then

$$t_q = m\ell + \tau_0(\beta, \ell) + \frac{\tau_1(\beta, \ell)}{m} + O\left(\frac{(1+\ell)^3}{m^2}\right), \quad (17.3)$$

where

$$\tau_0(\beta, \ell) = -\frac{1+\beta}{2}\ell - 1 - \frac{5\beta}{6}, \quad (17.4)$$

$$\tau_1(\beta, \ell) = \beta^2 \left(\frac{\ell^2}{8} + \frac{\ell}{3} + \frac{3}{8} \right) + \beta \left(\frac{\ell^2}{4} + \frac{\ell}{3} + \frac{5}{12} \right) - \frac{\ell}{12}. \quad (17.5)$$

More generally, for every fixed $J \geq 1$ there are effectively computable polynomials $\tau_0, \dots, \tau_{J-1} \in \mathbb{Q}[\beta, \ell]$ such that

$$t_q = m\ell + \sum_{h=0}^{J-1} \frac{\tau_h(\beta, \ell)}{m^h} + O_J\left(\frac{(1+\ell)^{J+1}}{m^J}\right). \quad (17.6)$$

The error is uniform over the stated compact set of target probabilities.

Proof. The critical-window expansion holds for the real exponent t , by theorem 15.1. It first locates the root at $t = m\ell + O(\log m)$: moving t/m by a fixed small constant on either side of ℓ moves the limiting value of $\log R_m(t)$ to either side of β , and the first correction then refines the displacement to $O(\log m)$.

Put $\varepsilon = 1/m$ and write $r = t/m = \ell + \tau_0\varepsilon + \tau_1\varepsilon^2 + \dots$. Then $\lambda = \beta \exp[-(r - \ell)]$. Insert these expressions into

$$\log R_m(t) = \lambda + \varepsilon L_1(\lambda, r) + \varepsilon^2 L_2(\lambda, r) + \dots$$

and set the result equal to β . The coefficient of ε gives

$$\tau_0 = \frac{L_1(\beta, \ell)}{\beta},$$

which is (17.4). The next coefficient gives

$$\tau_1 = \frac{\tau_0^2}{2} + \frac{\tau_0}{\beta} (\partial_r L_1 - \beta \partial_\lambda L_1)(\beta, \ell) + \frac{L_2(\beta, \ell)}{\beta}. \quad (17.7)$$

Simplification using (16.5)–(16.6) gives (17.5).

For the general construction, each $c_h(0, r)$ and $L_h(0, r)$ vanishes for $h \geq 1$, because $p_h(0, r) = 0$. Define polynomial coefficients $\mathcal{M}_h(\lambda, r)$ by

$$\log \left(1 + \sum_{h \geq 1} \frac{L_h(\lambda, r)}{\lambda} \varepsilon^h \right) = \sum_{h \geq 1} \mathcal{M}_h(\lambda, r) \varepsilon^h. \quad (17.8)$$

After taking the logarithm of $\log R_m(t) = \beta$, the formal inverse equation becomes

$$r = \ell + \sum_{h \geq 1} \mathcal{M}_h(\beta e^{-(r-\ell)}, r) \varepsilon^h. \quad (17.9)$$

This is triangular in powers of ε : its coefficient of order h determines the next coefficient of r using only earlier ones. It uses only polynomial operations and finite exponential series, so the resulting τ_h lie in $\mathbb{Q}[\beta, \ell]$. The degree bounds inherited from (15.8) show inductively that τ_h has degree at most $h + 1$ in ℓ .

[write] Added 5 October 2026, batch 98: proof of the degree bound. The manuscript only sketches this induction, and the error term $(1 + \ell)^{J+1}$ of (17.6) relies on it. A proof: by (15.8), $\deg_r c_h \leq h$; by induction on (16.3), L_h is a sum of products $c_{h_1} \cdots c_{h_k}$ with $h_1 + \cdots + h_k = h$, so $\deg_r L_h \leq h$; each L_h/λ is a polynomial (as $L_h(0, r) = 0$), and by (17.8) $\mathcal{M}_h$ is a sum of products of $L_{h_1}/\lambda, \dots, L_{h_k}/\lambda$ with $h_1 + \cdots + h_k = h$, so $\deg_r \mathcal{M}_h \leq h$. Let $\mathcal{A}$ be the set of formal series $\sum_{k \geq 1} P_k \varepsilon^k$ with $P_k \in \mathbb{Q}[\beta, \ell]$ and $\deg_\ell P_k \leq k$. It is closed under sums and products, it contains ε and $\varepsilon \ell$, and $e^{-i\omega} - 1 \in \mathcal{A}$ whenever $\omega \in \mathcal{A}$ and $i \geq 0$. For $\omega \in \mathcal{A}$ let $\Phi(\omega) = \sum_{h \geq 1} \mathcal{M}_h(\beta e^{-\omega}, \ell + \omega) \varepsilon^h$. A monomial $\lambda^i r^k$ of $\mathcal{M}_h$, $k \leq h$, contributes $\beta^i e^{-i\omega} \varepsilon^{h-k} (\varepsilon \ell + \varepsilon \omega)^k$, which lies in $\mathcal{A}$ (for $k = 0$ use $h \geq 1$); so $\Phi(\mathcal{A}) \subseteq \mathcal{A}$. Every term of $\Phi(\omega)$ carries a factor ε beyond the orders of ω it involves, so the coefficient of ε^k in $\Phi(\omega)$ depends only on the coefficients of ω of order below k . Hence the solution $\omega = r - \ell$ of (17.9) is the ε -adic limit of $\omega_0 = 0$, $\omega_{k+1} = \Phi(\omega_k)$, whose terms agree with ω through order k and lie in $\mathcal{A}$; so $\omega \in \mathcal{A}$. Since $\omega = \sum_{h \geq 0} \tau_h \varepsilon^{h+1}$, $\deg_\ell \tau_h \leq h + 1$. $\square$

A symbolic check of the write (formal inversion of the shipped $L_1, \dots, L_4$ of `data/02-poisson-defects-coefficients.json`, SymPy 1.14.0) reproduces (17.4)–(17.5) and finds $\deg_\ell \tau_h = \deg_\beta \tau_h = h + 1$ for $h = 0, 1, 2, 3$, with

$$\begin{aligned} \tau_2 = & -\frac{1}{6480} [(540\ell^3 + 1755\ell^2 + 2700\ell + 1601)\beta^3 + (1890\ell^3 + 3375\ell^2 + 4050\ell + 1665)\beta^2 \\ & + (540\ell^3 - 1350\ell^2 - 270\ell - 450)\beta + 270\ell]. \end{aligned}$$

At $q = e^{-1}$ it lowers the last column of table 2 from 1.38×10^{-4} to -1.5×10^{-6} at $m = 1000$ (and from 1.50×10^{-3} to -6.1×10^{-5} at $m = 200$). These are evidence for the computation, not part of the proof.

It remains to turn the formal calculation into a root estimate. A truncation through τ_{J-1} has a residual in $\log R_m(t) - \beta$ of size $O_J((1 + \ell)^{J+1} m^{-J-1})$, by the direct expansion and Taylor's theorem on a fixed compact critical window. On this window,

$$-\frac{d}{dt} \log R_m(t) = \frac{\sum_j d_j(m)(1 - j/m)^t [-\log(1 - j/m)]}{R_m(t)} \geq \frac{w_1}{m R_m(t)} \geq \frac{c_q}{m}$$

for a positive constant uniform over the target compact set. Here w_1 is bounded away from zero and $R_m(t)$ stays bounded on the critical window; at the root, $w_1 \rightarrow \beta$. The mean value theorem therefore multiplies the residual by at most $O(m)$, proving (17.6) and (17.3). $\square$

17.1 Integer thresholds and the rounding obstruction

For the original counting problem, the relevant threshold is

$$N_q(m) = \min\{n \in \mathbb{Z}_{\geq 0} : \mathbb{P}(D_{m,n} = 0) \geq q\} = \lceil t_q \rceil. \quad (17.10)$$

The equality is exact by strict monotonicity of $q_m(t)$. An asymptotic approximation to t_q with error tending to zero does not by itself justify taking its ceiling: the exact root may lie closer to an integer than the error. If a rigorously bounded error is η , the valid envelope is

$$\lceil \tilde{t}_q - \eta \rceil \leq N_q(m) \leq \lceil \tilde{t}_q + \eta \rceil.$$

For sufficiently large m the two sides differ by at most one if $\eta < 1/2$.

[write] **Added 5 October 2026, batch 98.** The qualifier “for sufficiently large m ” is superfluous: the two arguments differ by $2\eta < 1$, and the ceilings of two reals less than one apart differ by at most one, for every m . What needs large m , or explicit constants (question 20.6), is a rigorous $\eta < 1/2$.

The asymptotic constants in theorem 17.1 have not been optimized into a certified finite-input cutoff. For rational q , an exact comparison of integers in (10.1) decides any candidate boundary; for a nonrational specified target, a certified representation or interval evaluation of that target is additionally required.

18 A new local conclusion on the original OEIS diagonal

The boundary analysis concerns a two-parameter extension rather than a new OEIS identification. The local-limit result also gives an immediate conclusion for the original sequence itself. The prior report [3] defines positive constants x, y by

$$x = \log(1 + e^{-y}), \quad x = \frac{y}{1 + e^y}, \quad B = 2 - y + x, \quad (18.1)$$

and proves

$$\mathbb{E}K_{n,n} = \mu n + \mu_0 + O(n^{-1}), \quad \text{Var}(K_{n,n}) = \sigma^2 n + O(1), \quad (18.2)$$

where

$$\mu = \frac{1}{y} = 0.8737024332396683\dots, \quad \sigma^2 = \frac{B-1}{By^2} = 0.0889169868271170\dots,$$

and $\mu_0 = 0.1445656885777610\dots$. These numerical constants are quoted from the earlier manuscript and are not new numerical discoveries here.

Corollary 18.1 (Diagonal local limit theorem). *Let K_n have law*

$$\mathbb{P}(K_n = k) = \frac{k^n k! S(n, k)}{a_n}.$$

Then, uniformly for all integers k ,

$$\mathbb{P}(K_n = k) = \frac{1}{\sigma\sqrt{n}} \varphi\left(\frac{k - \mu n - \mu_0}{\sigma\sqrt{n}}\right) + O(n^{-1}). \quad (18.3)$$

All higher local corrections can be generated from exact cumulants by theorem 14.1; combining that theorem with the analytic marked expansion of [3] converts them to fixed diagonal constants.

Proof. On the diagonal, $\theta = (n-1)(1-1/n)^n/2 \sim n/(2e)$, hence the variance diverges. Apply (14.2) to $K_n = n - D$ using its exact mean and variance. Substitute (18.2); the change in the Gaussian density, uniformly in its argument, is $O(n^{-3/2})$ from the variance replacement and smaller from the stated mean replacement. This is absorbed by the $O(n^{-1})$ local error. $\square$

For clarity, the local theorem with exact moments does not depend on the earlier saddle proof. Only the explicit replacement of the moments by $\mu n + \mu_0$ and $\sigma^2 n$ invokes [3]. This separates the new finite-polynomial argument from inherited asymptotic input.

[write] **Added 5 October 2026, batch 98.** The proof establishes (18.3) only. The corollary’s last sentence, that the higher local corrections can be converted to fixed diagonal constants via Part I’s marked expansion (6.1), is a one-line sketch: Part I states (6.1) and (6.2) to first order only, and the bookkeeping that turns exact cumulants into constants is not carried out. It is recorded as open in question 20.8. This corollary answers Part I’s Question 4 in part (local limit theorem and, at exact cumulants, Edgeworth terms; no large deviations).

19 Reproducible checks and numerical stress tests

The package contains four short Python programs. The symbolic code uses rational arithmetic throughout; it does not estimate coefficients by fitting sequence values. The numerical code starts with exact integer ordered-Stirling rows and then uses high-precision arithmetic for normalization and noninteger exponent inversion.

19.1 Exact checks

The recurrence

$$T_{m,k} = k(T_{m-1,k} + T_{m-1,k-1}), \quad T_{0,0} = 1, \quad (19.1)$$

computes $T_{m,k} = k!S(m,k)$ using integers. The executed verification script checks the following.

1. Eighty-five coefficient identities: for each $j = 0, \dots, 16$, interpolate the degree- j polynomial $d_j(m)$ from independent exact Stirling values, multiply its normalized polynomial by the tilt series, and compare orders $0, \dots, 4$ with the general coefficient algorithm.
2. Fifty exact Sturm root counts for $m = 1, \dots, 10$ and $n \in \{0, 1, 2, 5, 20\}$, after dividing the block polynomial by u ; greatest-common-divisor checks also verify simplicity in these cases.
3. Two hundred seventy-two exact rational instances of the moment and Δ bounds, for $m = 2, \dots, 18$ and $n = 0, \dots, 15$. The first fifteen values of A122399 are also checked against the OEIS prefix displayed by the inspected entry.
4. The two printed inverse correction polynomials are substituted back into the direct logarithmic expansion, and the residual is verified to vanish through order m^{-2} by exact symbolic algebra.

All these checks passed. They test the implementation and the printed coefficients, not the infinite quantifiers in the analytic theorems. A separate standard-library program certifies one integer inverse threshold by exact counts and rational bounds, as explained below.

19.2 Boundary count accuracy

The following table uses n equal to the nearest integer to $m \log m$. An “error, J ” is

$$\left| R_{m,n} e^{-\lambda} - \sum_{h=0}^J c_h(\lambda, r) m^{-h} \right|.$$

Thus the errors refer to the normalized correction factor, not to the astronomically large integer count $A_{m,n}$.

m	n	$R_{m,n}$	error, $J = 0$	error, $J = 2$	error, $J = 4$
20	60	1.51840272	0.07708	0.003039	0.00019
50	196	1.57732508	0.0395	0.0003965	6.589×10^{-6}
100	461	1.60692912	0.023	7.793×10^{-5}	4.472×10^{-7}
200	1060	1.62582826	0.01306	1.438×10^{-5}	2.73×10^{-8}
500	3107	1.63933008	0.005998	1.426×10^{-6}	5.957×10^{-10}
1000	6908	1.64312456	0.003273	2.368×10^{-7}	3.047×10^{-11}

Table 1: Boundary expansion errors computed with 80-decimal-digit arithmetic.

At $(m, n) = (1000, 6908)$, the distance from the defect law to $\text{Poisson}(\theta)$ is approximately 9.61187×10^{-4} , while the explicit upper bound 2Δ is approximately 4.23702×10^{-3} . The latter is a theorem with an exact parameter; the displayed decimal evaluations are numerical diagnostics.

The delivered CSV also records scaled remainders $\text{error}(m/(1+r))^{J+1}$ for $J = 0, \dots, 4$. They remain of moderate size across the tested rows, consistent with the proved logarithmic remainder scale. This numerical observation is not used to infer the theorem’s constants.

19.3 Inverse accuracy

In table 2, the target probability is $q = e^{-1}$, so $\beta = 1$. The three errors are respectively $m\ell - t_q$, $m\ell + \tau_0 - t_q$, and $m\ell + \tau_0 + \tau_1/m - t_q$.

m	t_q	leading error	first correction error	second correction error
20	42.09797056	3.95373	-0.182187	0.0239655
50	156.01364645	4.93014	-0.122064	0.0090312
100	385.54124574	5.66105	-0.084302	0.0038250
200	914.65118974	6.38285	-0.055656	0.0014982
500	2753.40615791	7.32430	-0.030493	0.0003967
1000	6206.57891893	8.02918	-0.018762	0.0001379

Table 2: Inverting the exact real-exponent finite sum at target $q = e^{-1}$.

For $m = 1000$, the root of the exact real-exponent finite sum is numerically 6206.578918934... A separate exact-arithmetic certificate proves that the corresponding integer threshold is 6207. The leading approximation misses by about 8.03; adding the first correction reduces the error to about -0.0188 , and the second to 1.38×10^{-4} . The root finder uses the exact finite sum as its defining function, not the asymptotic approximation. Its high precision is not a rigorous interval certificate. The integer threshold does not rely on that root finder: put

$$E_- = \sum_{k=0}^{30} \frac{1}{k!}, \quad E_+ = E_- + \frac{32}{31 \cdot 31!}.$$

The exponential series and a geometric tail bound give $E_- < e < E_+$. The program `code/certify_threshold.py` computes the integer numerators and denominators of $R_{1000,6206}$ and $R_{1000,6207}$, and checks by cross multiplication that

$$R_{1000,6206} > E_+ > e > E_- > R_{1000,6207}.$$

Strict monotonicity then proves $N_{1/e}(1000) = 6207$ with no floating-point assumption. The rational bounds and comparison signs are saved in `data/threshold_certificate.json`.

19.4 Diagonal local checks

At $m = n = 1000$, the numerically evaluated defect mean and variance are

$$M = 126.15300932902\dots, \quad V = 88.84323982565\dots$$

The maximum absolute mass error over the entire finite support is about 4.66368×10^{-4} for the Gaussian approximation with exact moments, and 2.66632×10^{-5} after the first Edgeworth correction. The CSV records the corresponding errors for $m = n \in \{20, 50, 100, 200, 500, 1000\}$, as well as the scaled quantities $V \text{ error}_0$ and $V^{3/2} \text{ error}_1$. This directly tests the uniform absolute-error interpretation in theorem 14.1, rather than a central-window-only fit.

19.5 Files and commands

From the package directory, run

```
python code/coefficients.py --order 4
python code/verify.py
python code/numerics.py --dps 80
python code/certify_threshold.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The delivered run used Python 3, SymPy 1.14.0, and mpmath 1.3.0; the full runtime version is in `data/numerical_environment.json`. The numerical cases were independently rerun at 120 decimal digits; all stored 14-significant-digit diagnostics agreed, excluding the precision-dependent root residuals. The comparison is recorded in `data/precision_audit.json`. This is a precision audit, not an interval-arithmetic proof. The checked-in coefficient tables, numerical CSVs, and JSON summaries allow comparison with a rerun. The algorithm is intentionally transparent rather than optimized for very high expansion orders. In particular, the displayed fourth-order output is a reproducible example, not a claim that an exponential-size symbolic expression can be produced in polynomial time in its order.

[write] **Added 5 October 2026, batch 98: shipped names and the replay.** The package is shipped in this report's `code/` and `data/` with the prefix `02-poisson-defects-` (for example `code/02-poisson-defects-verify.py`, `data/02-poisson-defects-threshold_certificate.json`); its `Makefile` and `requirements.txt` are `code/02-poisson-defects-Makefile` and `data/02-poisson-defects-requirements.txt`, and its `SOURCE_AUDIT.md` is `02-poisson-defects-SOURCE_AUDIT.md`. The programs import one another and write `data/` by their delivered names, so the commands above work only in a copy with the delivered layout; the README of this report gives that route. In such a copy the write reran all four programs on 5 October 2026 (Windows, CPython 3.13.5, SymPy 1.14.0, mpmath 1.3.0): all passed, in 5, 170, 19 and 2 s for `coefficients.py`, `verify.py`, `numerics.py` and `certify_threshold.py`; the three CSV files came out byte-identical, the other outputs identical up to line endings, except the Python version string of `numerical_environment.json`. *The precision audit.* No shipped program writes `data/02-poisson-defects-precision_audit.json`; the comparison program behind it was not delivered, so the file is a record, not a reproducible output. The write repeated the comparison with its own script: `numerics.py--dps120` in a separate copy, compared field by field with the shipped 80-digit CSV files, gives 306, 84 and 42 equal fields (432 in all) besides the 12 root residuals, which differ in 11 rows, as the record says.

20 Further questions and research

The questions below are left open by this article. Some may admit answers from broader existing literature; the wording does not claim that each is a newly posed or universally unresolved problem.

[write] **Added 5 October 2026, batch 98.** The manuscript titles this section “Further research questions”; its six questions are Questions 20.1–20.6 below, unchanged. Under the repository’s standing rule (4 October 2026) the write adds Questions 20.7 and 20.8, for two statements of the manuscript that are sketched or not asserted rather than proved, and a dated note at Question 20.4. The manuscript’s sketched degree bound $\deg_{\ell} \tau_h \leq h + 1$ is not listed: the write proves it in section 17. No statement of the manuscript was found to be false.

Question 20.1 (The sharp Poisson window). Determine the widest regime in which the defect law is close in total variation to a Poisson law with its exact mean. The bound 2Δ is simple and explicit but loses a possible large-mean improvement. Can one give necessary and sufficient conditions in terms of m, n , and obtain a sharp leading constant for the distance?

A plausible route is to combine sharper Bernoulli–Poisson estimates with asymptotics of the root power sums $\sum_i \alpha_i^a$. This must be kept separate from the stronger requirement $R_{m,n}/e^{\lambda} \rightarrow 1$, which concerns the count normalization.

Question 20.2 (The connected-degree pattern). The generated logarithmic coefficients satisfy $\deg_{\lambda} L_h = h + 1$ for $1 \leq h \leq 4$, although the general theorem only proves the upper bound $2h$. Prove or refute

$$\deg_{\lambda} L_h \leq h + 1 \quad (h \geq 1)$$

and identify a connected combinatorial interpretation of their highest powers of λ .

Such a result could greatly enlarge the useful growing- λ window of the logarithmic expansion. Finite verification through order four is not a proof of the pattern.

Question 20.3 (A uniform bridge to the interior saddle). Construct a single count approximation with controlled relative error that connects the present finite-defect regime, the growing-defect Poisson region, and the fixed-positive-direction saddle expansion of [3]. Determine the correct rescaled special function, if any, when the Gaussian saddle approaches the endpoint.

The all-direction local theorem already controls the distribution at its exact moments; what remains here is explicit asymptotic evaluation of the count and of those moments across the whole transition.

Question 20.4 (The zero distribution and real exponent continuation). For $n/m \rightarrow r_0 \geq 0$, find a limiting empirical measure of the α_i and derive root-density formulas for the cumulants. Separately, classify real $t \geq 0$ for which $\sum_k k^t k! S(m, k) u^k$ remains real-rooted.

The proof by repeated Euler differentiation applies only to integer t . The inverse theorem does not assume an affirmative answer for real t , so this issue is an extension rather than a hidden dependency.

[write] **Added 5 October 2026, batch 98: evidence only.** For $t \in \{0.5, 1.5, 2.25, 3.3, 7.7\}$ and $m = 2, \dots, 20$ (95 polynomials), the write computed all zeros of $\sum_k k^t k! S(m, k) u^{k-1}$ with mpmath at 60 digits: every zero is real and lies in $(-1, 0)$. This is numerical evidence for an affirmative answer, not a proof; the script is not shipped.

Question 20.5 (Beyond every algebraic order). Determine the large-order growth of c_h and L_h , the possible optimal truncation scale, and the exponentially small sectors associated with noncentral complex singularities. Can the inverse expansion be made exponentially accurate with a corresponding Stokes analysis?

Nothing in the present fixed-order Cauchy argument supplies those conclusions. A solution should distinguish genuine complex-saddle contributions from artificial contour endpoints.

Question 20.6 (A formal and effective proof layer). Formalize the exact Stirling recurrences, root interlacing, Bernoulli factorization, and finite-input inequalities in Lean. Then replace the existence constants in the boundary and inverse remainder theorems by explicit computable bounds, sufficiently sharp to certify integer thresholds directly.

The finite part requires only polynomial algebra, Rolle’s theorem, finite probability, and elementary inequalities. It is a natural first formalization target before the all-order analytic estimates.

Question 20.7 ([write] The total-variation half of the growing window). Source: the last sentence of section 12, “No assertion that this sufficient window is sharp is made”. Let $\lambda \geq \lambda_0 > 0$ grow with m . Find necessary and sufficient conditions for $d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(\lambda)) \rightarrow 0$.

What is known: (12.14) gives sufficiency of $\lambda^2(1+r)/m \rightarrow 0$, and the write’s note in section 12 shows that this condition is *necessary* for the count statement $R_{m,n}e^{-\lambda} \rightarrow 1$. For the distance it is probably not: a heuristic sketch, not proved here, combines the Barbour–Hall bound [9] $d_{\text{TV}}(\mathcal{L}(D), \text{Poisson}(M)) \leq \Delta/M$ (up to the factor $1 - e^{-M}$), which is of order $\lambda(1+r)/m$, with the distance between $\text{Poisson}(M)$ and $\text{Poisson}(\lambda)$, of order $|\lambda - M|/\sqrt{\lambda} \lesssim \lambda^{3/2}(1+r)/m$ by (12.12) and $\theta - M \leq \Delta$. That suggests that $\lambda^{3/2}(1+r)/m \rightarrow 0$ suffices, and perhaps is necessary, since the mean mismatch is then larger than a standard deviation. Missing: a checked statement of the Barbour–Hall bound and of the Poisson–Poisson distance with constants, and a matching lower bound. This sharpens the question of the window in Question 20.1, which asks about the exact-mean Poisson law.

Question 20.8 ([write] Fixed-constant Edgeworth terms on the diagonal). Source: the last sentence of corollary 18.1, which asserts without proof that the higher local corrections of theorem 14.1 can be converted to fixed diagonal constants by Part I’s marked expansion. Prove this, and compute the first correction explicitly.

The source’s sketch: Part I extends its saddle expansion uniformly to a small complex disk of the marking variable s and allows its remainder to be differentiated by Cauchy estimates (its Section 6). Carried to every fixed order, this would give $\kappa_a(K_n) = n(\log D)^{(a)}(0) + (\log C)^{(a)}(0) + O(n^{-1})$ and finer expansions, with Part I’s $D(s)$ and $C(s)$; inserting them in (14.1) and re-expanding the Gaussian factor about $(k - \mu n - \mu_0)/(\sigma\sqrt{n})$ would express each correction through fixed constants. Missing: Part I states (6.1) and (6.2) only to first order and computes no derivative of $\log D(s)$ beyond the second; the all-order version in s and the bookkeeping are not written. Without them (18.3) is what is proved.

21 Conclusion

The same array that has a fixed-direction saddle expansion also has a simple endpoint geometry. The exact block polynomial exposes that geometry: all of its nonzero zeros lie in $(-1, 0)$, so missing blocks form an independent-Bernoulli sum. Two coefficients control its Poisson distance and count normalization, while the variance alone controls its local Gaussian expansion. At $n = m(\log m + c)$, finitely many pair defects survive; their Poisson parameter is $e^{-c}/2$.

The finite-defect identity then produces a complete, uniformly summable power–logarithmic expansion. Its inversion gives the number of blue labels needed for a prescribed all-singleton probability, including the logarithmic correction and every further algebraic order. This extends

the inspected ProveIt report in directions it explicitly left outside its scope, while retaining a clear boundary between proved fixed-order results, numerical diagnostics, inherited diagonal constants, and questions about sharp uniformity or beyond-all-orders behavior.

A The third logarithmic correction

For reference, the next coefficient in (16.4) is

$$\begin{aligned}
L_3(\lambda, r) = & \lambda \left(-\frac{r^3}{48} + \frac{r^2}{24} + \frac{r}{12} \right) \\
& + \lambda^2 \left(-\frac{7r^3}{12} - \frac{5r^2}{6} - \frac{13r}{36} \right) \\
& + \lambda^3 \left(-\frac{17r^3}{12} - \frac{23r^2}{6} - 4r - \frac{5}{3} \right) \\
& + \lambda^4 \left(-\frac{2r^3}{3} - \frac{7r^2}{3} - \frac{107r}{36} - \frac{251}{180} \right). \tag{A.1}
\end{aligned}$$

The direct third coefficient is

$$c_3 = L_3 + L_1 L_2 + \frac{L_1^3}{6}.$$

The accompanying `data/coefficients.txt` and `data/coefficients.json` include p_h, c_h, L_h through $h = 4$. They are generated by the recurrence in section 15, rather than manually entered expressions.

B Source and dependency audit

The repository tree inspected during preparation had commit

`0f93381e8932d3aee3d5c2abfb5c98d128d49b9c`.

The inspected `proof.md` file has Git blob SHA

`6f03a8be2d313d67cebce8a5f5f641bbb78e08d1`.

Its fixed-direction theorem and its stated absence of local and boundary-uniform claims motivated the choice of topic. The finite identities, root theorem, global bounds, boundary expansion, local theorem with exact cumulants, and inverse-threshold theorem in the present article are proved independently of the earlier analytic saddle argument. The explicit diagonal constants in section 18 are the only inherited analytic inputs to a numbered consequence here.

OEIS A122399 and A317855 were inspected on 4 October 2026. The classical papers cited below provide provenance for methods and neighboring problems; none is invoked as a substitute for a missing proof of the specific boundary formulas. Targeted searches are recorded in `SOURCE_AUDIT.md`. The manuscript is neither a submission to OEIS nor a report of independent mathematical refereeing. [write] **Added 5 October 2026, batch 98.** The commit `0f93381e8` is an ancestor of the placement commit, and `6f03a8be2` is still the blob of this report’s `proof.md`; the report had no commit between them, so the manuscript read the current Part I. `SOURCE_AUDIT.md` is shipped as `02-poisson-defects-SOURCE_AUDIT.md`. Its scoped search (“A122399 Poisson”) was right that this report contained no defect–Poisson transition; the collection’s `a261781-matrix-compositions` has one for a different array (a Poisson limit for Stirling defects, its Theorem `mxc:tp:thm:defect`, and a Touchard-summed coverage-window expansion to every order, its Theorem `mxc:tp:thm:couponall`), with the same mechanism of pairs weighted by $1/2!$ but no shared theorem. The three items the write added to the bibliography, [7, 8, 9], are cited for priority and context; the write did not consult them.

References for Part II

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A122399, “ $a(n) = \sum_{k=0}^n k^n k! \text{Stirling2}(n, k)$,” <https://oeis.org/A122399>. Inspected 4 October 2026.
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A317855, the growth constant associated with A122399, <https://oeis.org/A317855>. Inspected 4 October 2026.
- [3] Part I of this report, as delivered: *A122399: a complete saddle expansion, an exact vertical contour, and quantitative inversion*, research manuscript dated 1 October 2026, `proof.md` in the A122399 report directory identified in section 9, of the ProveIt repository, <https://github.com/VladimirReshetnikov/ProveIt>. Inspected commit and blob are recorded in appendix B. **[write]** Corrected 5 October 2026, batch 98: the delivered item began “Vladimir Reshetnikov, ProveIt repository,” and added “This citation names the repository owner, not an inferred personal authorship of every statement in its report collection.” The manuscript it cites names no author (Part I’s provenance note), so no author is named here.
- [4] Lily L. Liu and Yi Wang, *A unified approach to polynomial sequences with only real zeros*, *Advances in Applied Mathematics* **38** (2007), 542–560; arXiv:math/0509207, <https://arxiv.org/abs/math/0509207>.
- [5] Lucien Le Cam, *An approximation theorem for the Poisson binomial distribution*, *Pacific Journal of Mathematics* **10** (1960), no. 4, 1181–1197. doi:10.2140/pjm.1960.10.1181. The elementary convolution estimate is in Theorem 1 and its proof, pp. 1183–1184.
- [6] Jessica Khera, Erik Lundberg, and Stephen Melczer, *Asymptotic enumeration of lonesum matrices*, arXiv:1912.08850, version 2 (2020), <https://arxiv.org/abs/1912.08850>. Published in *Advances in Applied Mathematics* **123** (2021), 102118.
- [7] **[write]** Georg Frobenius, *Über die Bernoullischen Zahlen und die Eulerschen Polynome*, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften* (1910), 809–847.
- [8] **[write]** L. H. Harper, *Stirling behavior is asymptotically normal*, *Annals of Mathematical Statistics* **38** (1967), 410–414.
- [9] **[write]** A. D. Barbour and Peter Hall, *On the rate of Poisson convergence*, *Mathematical Proceedings of the Cambridge Philosophical Society* **95** (1984), 473–480.